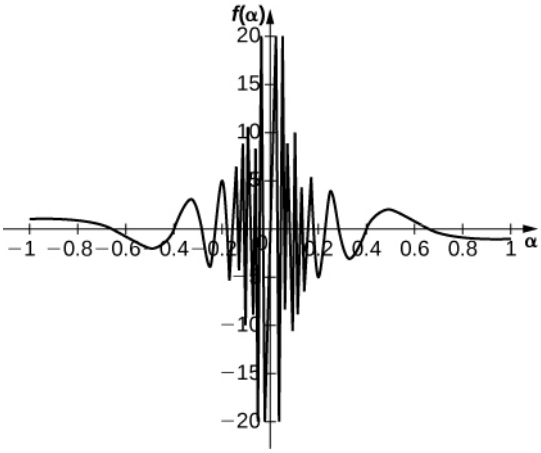


47. False; $\lim_{x \to -2^+} f(x) = +\infty$

49. False; $\lim_{x \to 6} f(x)$ DNE since $\lim_{x \to 6^-} f(x) = 2$ and $\lim_{x \to 6^+} f(x) = 5$.

51. 2

53. 1

55. 1

57. DNE

59. 0

61. DNE

63. 2

65. 3

67. DNE

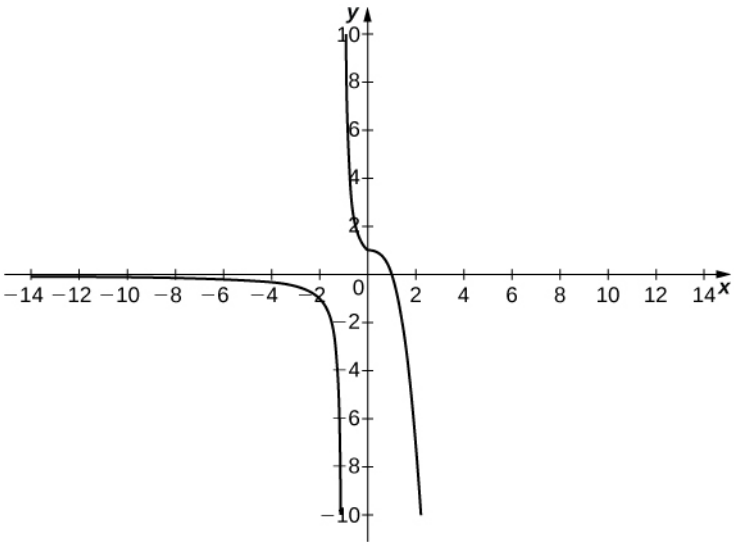
69. 0

71. −2

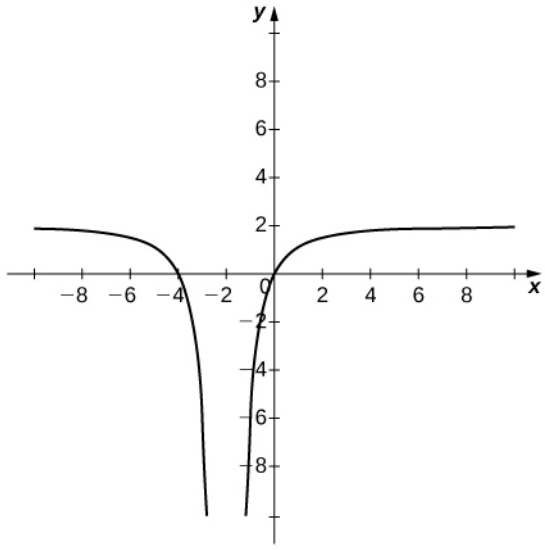
73. DNE

75. 0

77. Answers may vary.



79. Answers may vary.



81. a. ρ_2 b. ρ_1 c. DNE unless $\rho_1 = \rho_2$. As you approach x_{SF} from the left, you are in the high-density area of the shock. When you approach from the right, you have not experienced the "shock" yet and are at a lower density.

Section 2.3 Exercises

83. Use constant multiple law and difference law: $\lim\limits_{x\to 0}\left(4x^2-2x+3\right)=4\lim\limits_{x\to 0}x^2-2\lim\limits_{x\to 0}x+\lim\limits_{x\to 0}3=3$

85. Use root law: $\lim\limits_{x\to -2}\sqrt{x^2-6x+3}=\sqrt{\lim\limits_{x\to -2}\left(x^2-6x+3\right)}=\sqrt{19}$

87. 49

89. 1

91. $-\frac{5}{7}$

93. $\lim\limits_{x\to 4}\frac{x^2-16}{x-4}=\frac{16-16}{4-4}=\frac{0}{0}$; then, $\lim\limits_{x\to 4}\frac{x^2-16}{x-4}=\lim\limits_{x\to 4}\frac{(x+4)(x-4)}{x-4}=8$

95. $\lim\limits_{x\to 6}\frac{3x-18}{2x-12}=\frac{18-18}{12-12}=\frac{0}{0}$; then, $\lim\limits_{x\to 6}\frac{3x-18}{2x-12}=\lim\limits_{x\to 6}\frac{3(x-6)}{2(x-6)}=\frac{3}{2}$

97. $\lim\limits_{x\to 9}\frac{t-9}{\sqrt{t}-3}=\frac{9-9}{3-3}=\frac{0}{0}$; then, $\lim\limits_{t\to 9}\frac{t-9}{\sqrt{t}-3}=\lim\limits_{t\to 9}\frac{t-9}{\sqrt{t}-3}\frac{\sqrt{t}+3}{\sqrt{t}+3}=\lim\limits_{t\to 9}\left(\sqrt{t}+3\right)=6$

99. $\lim\limits_{\theta\to\pi}\frac{\sin\theta}{\tan\theta}=\frac{\sin\pi}{\tan\pi}=\frac{0}{0}$; then, $\lim\limits_{\theta\to\pi}\frac{\sin\theta}{\tan\theta}=\lim\limits_{\theta\to\pi}\frac{\sin\theta}{\frac{\sin\theta}{\cos\theta}}=\lim\limits_{\theta\to\pi}\cos\theta=-1$

101. $\lim\limits_{x\to 1/2}\frac{2x^2+3x-2}{2x-1}=\frac{\frac{1}{2}+\frac{3}{2}-2}{1-1}=\frac{0}{0}$; then, $\lim\limits_{x\to 1/2}\frac{2x^2+3x-2}{2x-1}=\lim\limits_{x\to 1/2}\frac{(2x-1)(x+2)}{2x-1}=\frac{5}{2}$

103. $-\infty$

105. $-\infty$

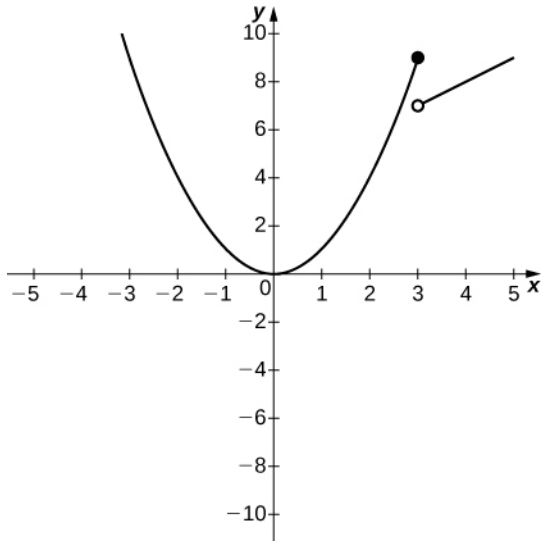
107. $\lim\limits_{x\to 6}2f(x)g(x)=2\lim\limits_{x\to 6}f(x)\lim\limits_{x\to 6}g(x)=72$

109. $\lim\limits_{x\to 6}\left(f(x)+\frac{1}{3}g(x)\right)=\lim\limits_{x\to 6}f(x)+\frac{1}{3}\lim\limits_{x\to 6}g(x)=7$

111. $\lim\limits_{x\to 6}\sqrt{g(x)-f(x)}=\sqrt{\lim\limits_{x\to 6}g(x)-\lim\limits_{x\to 6}f(x)}=\sqrt{5}$

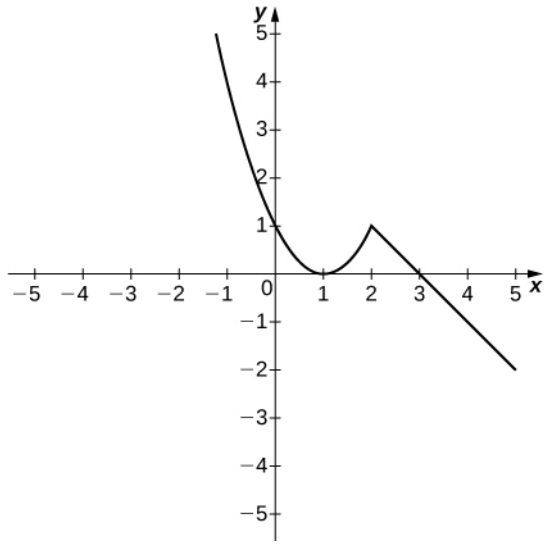
113. $\lim\limits_{x\to 6}[(x+1)f(x)]=\left(\lim\limits_{x\to 6}(x+1)\right)\left(\lim\limits_{x\to 6}f(x)\right)=28$

115.



a. 9; b. 7

117.



a. 1; b. 1

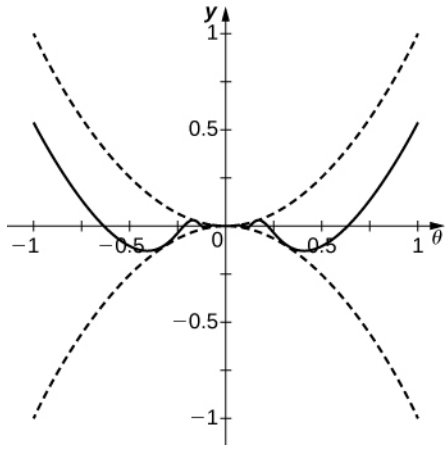
119. $\lim\limits_{x\to -3^-}(f(x)-3g(x)) = \lim\limits_{x\to -3^-} f(x) - 3\lim\limits_{x\to -3^-} g(x) = 0+6=6$

121. $\lim\limits_{x\to -5}\frac{2+g(x)}{f(x)} = \frac{2+\left(\lim\limits_{x\to -5} g(x)\right)}{\lim\limits_{x\to -5} f(x)} = \frac{2+0}{2} = 1$

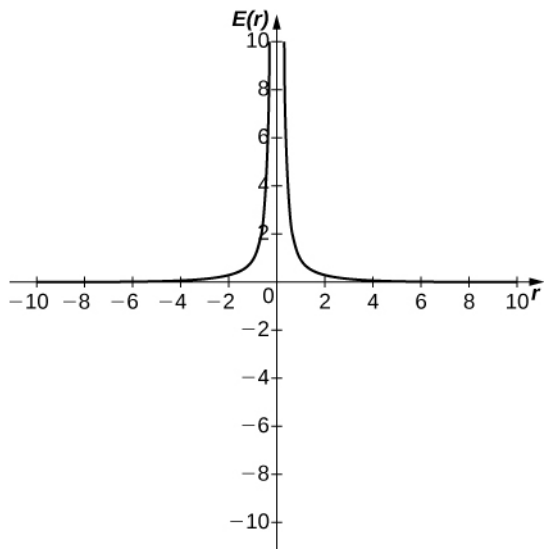
123. $\lim\limits_{x\to 1}\sqrt[3]{f(x)-g(x)} = \sqrt[3]{\lim\limits_{x\to 1} f(x) - \lim\limits_{x\to 1} g(x)} = \sqrt[3]{2+5} = \sqrt[3]{7}$

125. $\lim\limits_{x\to -9}(xf(x)+2g(x)) = \left(\lim\limits_{x\to -9} x\right)\left(\lim\limits_{x\to -9} f(x)\right) + 2\lim\limits_{x\to -9}(g(x)) = (-9)(6)+2(4) = -46$

127. The limit is zero.



129. a.



b. ∞. The magnitude of the electric field as you approach the particle *q* becomes infinite. It does not make physical sense to evaluate negative distance.

Section 2.4 Exercises

131. The function is defined for all *x* in the interval $\left(0, \infty\right)$.

133. Removable discontinuity at $x = 0$; infinite discontinuity at $x = 1$

135. Infinite discontinuity at $x = \ln 2$

137. Infinite discontinuities at $x = \frac{(2k+1)\pi}{4}$, for $k = 0, \pm 1, \pm 2, \pm 3, \ldots$

139. No. It is a removable discontinuity.

141. Yes. It is continuous.

143. Yes. It is continuous.

145. $k = -5$

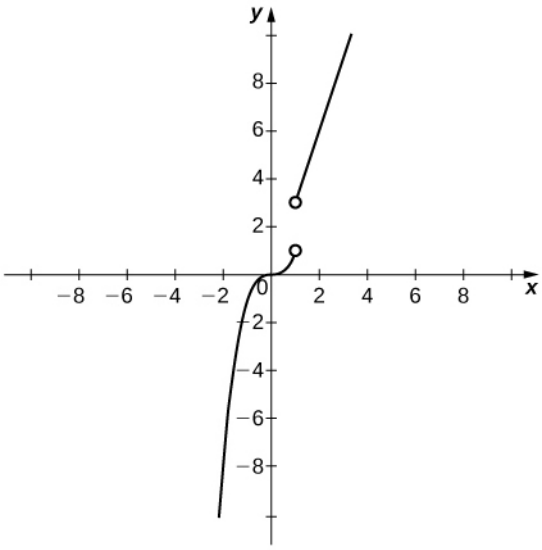
147. $k = -1$

149. $k = \frac{16}{3}$

151. Since both *s* and $y = t$ are continuous everywhere, then $h(t) = s(t) - t$ is continuous everywhere and, in particular, it is continuous over the closed interval $[2, 5]$. Also, $h(2) = 3 > 0$ and $h(5) = -3 < 0$. Therefore, by the IVT, there is a value $x = c$ such that $h(c) = 0$.

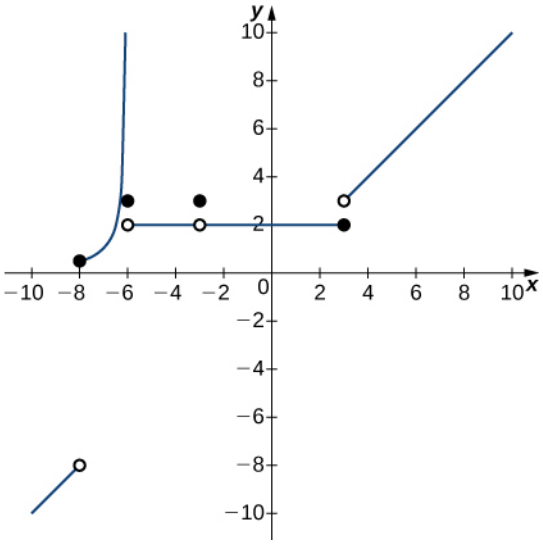
153. The function $f(x) = 2^x - x^3$ is continuous over the interval $[1.25, 1.375]$ and has opposite signs at the endpoints.

155. a.

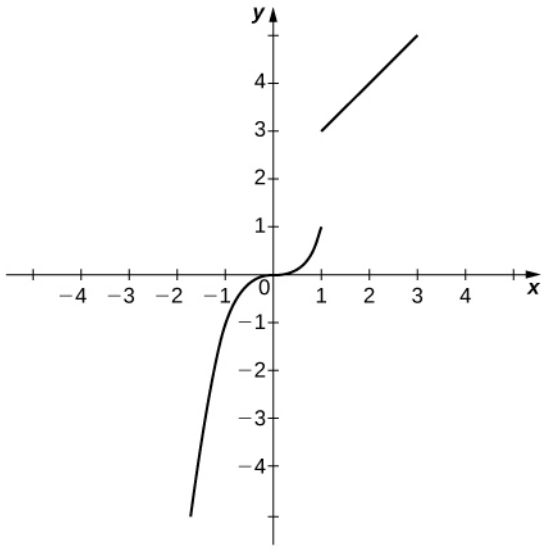


b. It is not possible to redefine $f(1)$ since the discontinuity is a jump discontinuity.

157. Answers may vary; see the following example:



159. Answers may vary; see the following example:



161. False. It is continuous over $\left(-\infty, 0\right) \cup \left(0, \infty\right)$.

163. False. Consider $f(x) = \begin{cases} x \text{ if } x \neq 0 \\ 4 \text{ if } x = 0 \end{cases}$.

165. False. IVT only says that there is at least one solution; it does not guarantee that there is exactly one. Consider $f(x) = \cos(x)$ on $[-\pi, 2\pi]$.

167. False. The IVT does *not* work in reverse! Consider $(x-1)^2$ over the interval $[-2, 2]$.

169. $R = 0.0001519 \text{ m}$

171. $D = 345{,}826 \text{ km}$

173. For all values of a, $f(a)$ is defined, $\lim\limits_{\theta \to a} f(\theta)$ exists, and $\lim\limits_{\theta \to a} f(\theta) = f(a)$. Therefore, $f(\theta)$ is continuous everywhere.

175. Nowhere

Section 2.5 Exercises

177. For every $\varepsilon > 0$, there exists a $\delta > 0$, so that if $0 < |t - b| < \delta$, then $|g(t) - M| < \varepsilon$

179. For every $\varepsilon > 0$, there exists a $\delta > 0$, so that if $0 < |x - a| < \delta$, then $|\varphi(x) - A| < \varepsilon$

181. $\delta \leq 0.25$

183. $\delta \leq 2$

185. $\delta \leq 1$

187. $\delta < 0.3900$

189. Let $\delta = \varepsilon$. If $0 < |x - 3| < \varepsilon$, then $|x + 3 - 6| = |x - 3| < \varepsilon$.

191. Let $\delta = \sqrt[4]{\varepsilon}$. If $0 < |x| < \sqrt[4]{\varepsilon}$, then $\left|x^4\right| = x^4 < \varepsilon$.

193. Let $\delta = \varepsilon^2$. If $5 - \varepsilon^2 < x < 5$, then $\left|\sqrt{5 - x}\right| = \sqrt{5 - x} < \varepsilon$.

195. Let $\delta = \varepsilon/5$. If $1 - \varepsilon/5 < x < 1$, then $|f(x) - 3| = 5x - 5 < \varepsilon$.

197. Let $\delta = \sqrt{\frac{3}{M}}$. If $0 < |x + 1| < \sqrt{\frac{3}{M}}$, then $f(x) = \frac{3}{(x+1)^2} > M$.

199. The engineer must cut within 0.328 cm of 12 cm on each side; $\varepsilon = 8, \delta = 0.328, a = 12, L = 144$

201. Answers may vary.

203. 0

205. $\lim\limits_{x \to a}(f(x)) + \lim\limits_{x \to a}(g(x))$
$= L + M$

207. Answers may vary.

Review Exercises

209. False

211. False. A removable discontinuity is possible.

213. 5

215. 8/7
217. DNE
219. 2/3
221. −4;
223. Since $-1 \le \cos(2\pi x) \le 1$, then $-x^2 \le x^2\cos(2\pi x) \le x^2$. Since $\lim\limits_{x\to 0} x^2 = 0 = \lim\limits_{x\to 0} -x^2$, it follows that $\lim\limits_{x\to 0} x^2\cos(2\pi x) = 0$.
225. $[2, \infty]$
227. $c = -1$
229. $\delta = \sqrt[3]{\varepsilon}$
231. 0 m/sec

Chapter 3

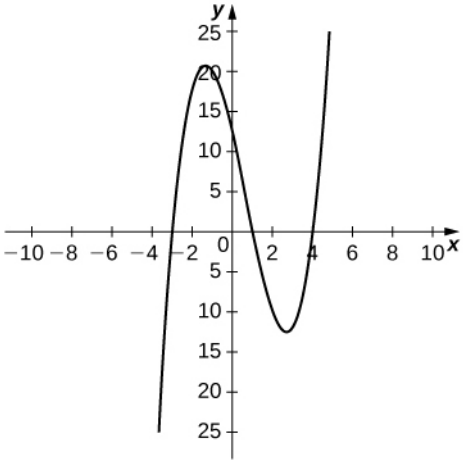
Checkpoint

3.1 $\frac{1}{4}$
3.2 6
3.3 $f'(1) = 5$
3.4 −32 ft/s
3.5 $P'(3.25) = 20 > 0$; raise prices
3.6 $f'(x) = 2x$
3.7 $\left(0, +\infty\right)$
3.8 $a = 6$ and $b = -9$
3.9 $f''(x) = 2$
3.10 $a(t) = 6t$
3.11 0
3.12 $4x^3$
3.13 $f'(x) = 7x^6$
3.14 $f'(x) = 6x^2 - 12x.$
3.15 $y = 12x - 23$
3.16 $j'(x) = 10x^4\left(4x^2 + x\right) + (8x+1)\left(2x^5\right) = 56x^6 + 12x^5.$
3.17 $k'(x) = -\frac{13}{(4x-3)^2}.$
3.18 $g'(x) = -7x^{-8}.$
3.19 $3f'(x) - 2g'(x).$
3.20 $\frac{5}{8}$
3.21 −4.4
3.22 left to right
3.23 3,300
3.24 $2
3.25 $f'(x) = \cos^2 x - \sin^2 x$
3.26 $\frac{\cos x + x\sin x}{\cos^2 x}$
3.27 $t = \frac{\pi}{3}, t = \frac{2\pi}{3}$
3.28 $f'(x) = -\csc^2 x$
3.29 $f'(x) = 2\sec^2 x + 3\csc^2 x$
3.30 $\frac{4}{3}$
3.31 $\cos x$
3.32 $-\cos x$
3.33 $v\left(\frac{5\pi}{6}\right) = -\sqrt{3} < 0$ and $a\left(\frac{5\pi}{6}\right) = -1 < 0$. The block is speeding up.
3.34 $h'(x) = 4\left(2x^3 + 2x - 1\right)^3\left(6x^2 + 2\right) = 8\left(3x^2 + 1\right)\left(2x^3 + 2x - 1\right)^3$
3.35 $y = -48x - 88$
3.36 $h'(x) = 7\cos(7x + 2)$
3.37 $h'(x) = \frac{3-4x}{(2x+3)^4}$
3.38 $h'(x) = 18x^2\sin^5\left(x^3\right)\cos\left(x^3\right)$

3.39 $a(t) = -16\sin(4t)$
3.40 28
3.41 $\frac{dy}{dx} = -3x^2 \sin\left(x^3\right)$
3.42 $g'(x) = -\frac{1}{(x+2)^2}$
3.43 $g(x) = \frac{1}{5}x^{-4/5}$
3.44 $s'(t) = (2t+1)^{-1/2}$
3.45 $g'(x) = \frac{1}{1+x^2}$
3.46 $h'(x) = \frac{-3}{\sqrt{6x-9x^2}}$
3.47 $y = x$
3.48 $\frac{dy}{dx} = \frac{5-20x^4}{\sec^2 y - 2y}$
3.49 $y = \frac{5}{3}x - \frac{16}{3}$
3.50 $h'(x) = e^{2x} + 2xe^{2x}$
3.51 996
3.52 $f'(x) = \frac{15}{3x+2}$
3.53 $9\ln(3)$
3.54 $\frac{dy}{dx} = x^x(1+\ln x)$
3.55 $y' = \pi(\tan x)^{\pi-1}\sec^2 x$

Section 3.1 Exercises

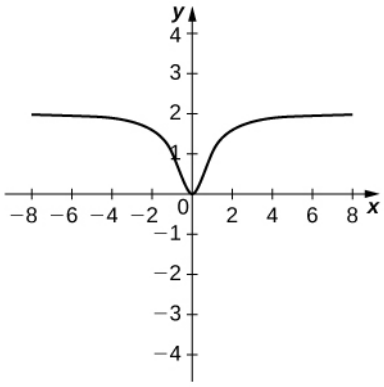
1. 4
3. 8.5
5. $-\frac{3}{4}$
7. 0.2
9. 0.25
11. a. -4 b. $y = 3 - 4x$
13. a. 3 b. $y = 3x - 1$
15. a. $\frac{-7}{9}$ b. $y = \frac{-7}{9}x + \frac{14}{3}$
17. a. 12 b. $y = 12x + 14$
19. a. -2 b. $y = -2x - 10$
21. 5
23. 13
25. $\frac{1}{4}$
27. $-\frac{1}{4}$
29. -3
31. a. (i) 5.100000, (ii) 5.010000, (iii) 5.001000, (iv) 5.000100, (v) 5.000010, (vi) 5.000001, (vii) 4.900000, (viii) 4.990000, (ix) 4.999000, (x) 4.999900, (xi) 4.999990, (x) 4.999999 b. $m_{\tan} = 5$ c. $y = 5x + 3$
33. a. (i) 4.8771, (ii) 4.9875 (iii) 4.9988, (iv) 4.9999, (v) 4.9999, (vi) 4.9999 b. $m_{\tan} = 5$ c. $y = 5x + 10$
35. a. $\frac{1}{3}$; b. (i) $0.\overline{3}$ m/s, (ii) $0.\overline{3}$ m/s, (iii) $0.\overline{3}$ m/s, (iv) $0.\overline{3}$ m/s; c. $0.\overline{3} = \frac{1}{3}$ m/s
37. a. $2\left(h^2 + 6h + 12\right)$; b. (i) 25.22 m/s, (ii) 24.12 m/s, (iii) 24.01 m/s, (iv) 24 m/s; c. 24 m/s
39. a. 1.25; b. 0.5
41. $\lim\limits_{x\to 0^-} \frac{x^{1/3}-0}{x-0} = \lim\limits_{x\to 0^-} \frac{1}{x^{2/3}} = \infty$
43. $\lim\limits_{x\to 1^-} \frac{1-1}{x-1} = 0 \neq 1 = \lim\limits_{x\to 1^+} \frac{x-1}{x-1}$
45. a. (i) 61.7244 ft/s, (ii) 61.0725 ft/s (iii) 61.0072 ft/s (iv) 61.0007 ft/s b. At 4 seconds the race car is traveling at a rate/velocity of 61 ft/s.
47. a. The vehicle represented by $f(t)$, because it has traveled 2 feet, whereas $g(t)$ has traveled 1 foot. b. The velocity of $f(t)$ is constant at 1 ft/s, while the velocity of $g(t)$ is approximately 2 ft/s. c. The vehicle represented by $g(t)$, with a velocity of approximately 4 ft/s. d. Both have traveled 4 feet in 4 seconds.
49. a.



b. $a \approx -1.361, 2.694$

51. a. $N(x) = \frac{x}{30}$ b. ~ 3.3 gallons. When the vehicle travels 100 miles, it has used 3.3 gallons of gas. c. $\frac{1}{30}$. The rate of gas consumption in gallons per mile that the vehicle is achieving after having traveled 100 miles.

53. a.



b. $-0.028, -0.16, 0.16, 0.028$

Section 3.2 Exercises

55. -3

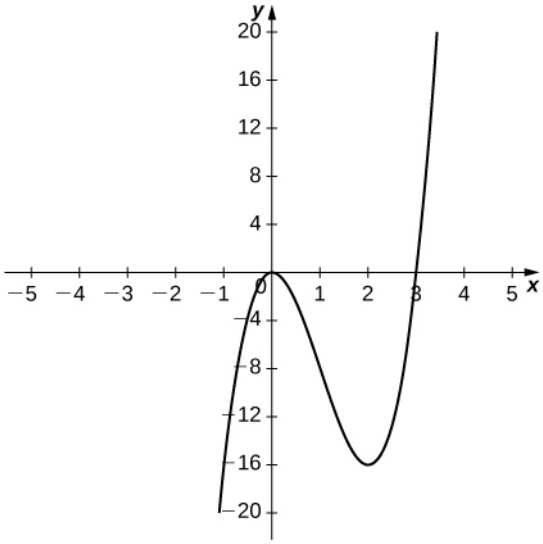
57. $8x$

59. $\frac{1}{\sqrt{2x}}$

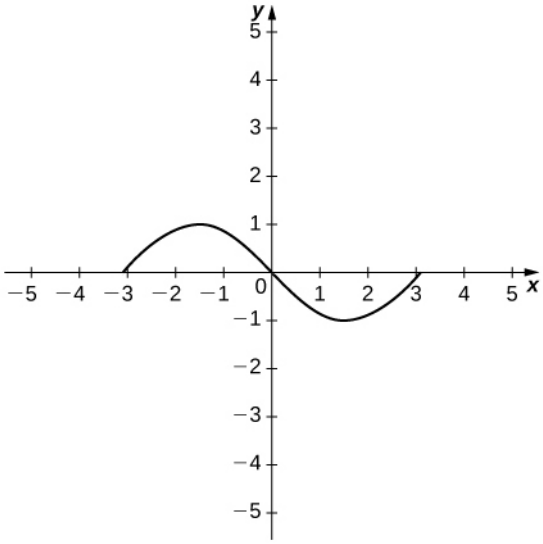
61. $\frac{-9}{x^2}$

63. $\frac{-1}{2x^{3/2}}$

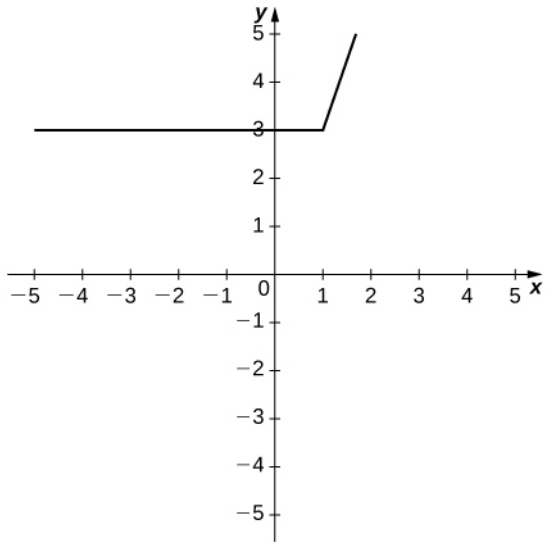
65.



67.

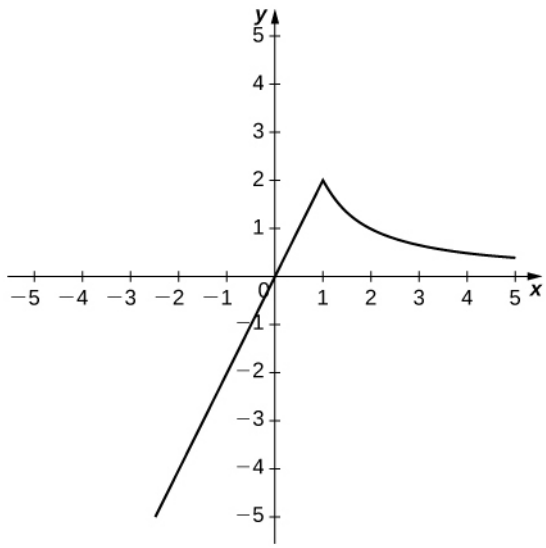


69. $f(x) = 3x^2 + 2, a = 2$
71. $f(x) = x^4, a = 2$
73. $f(x) = e^x, a = 0$
75. a.



b. $\lim\limits_{h \to 1^-} \frac{3-3}{h} \neq \lim\limits_{h \to 1^+} \frac{3h}{h}$

77. a.



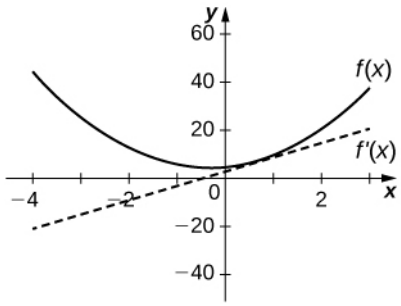
b. $\lim\limits_{h \to 1^-} \frac{2h}{h} \neq \lim\limits_{h \to 1^+} \frac{\frac{2}{x+h} - \frac{2}{x}}{h}$.

79. a. $x = 1$, b. $x = 2$

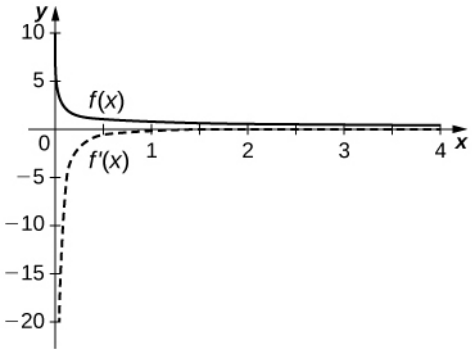
81. 0

83. $\frac{2}{x^3}$

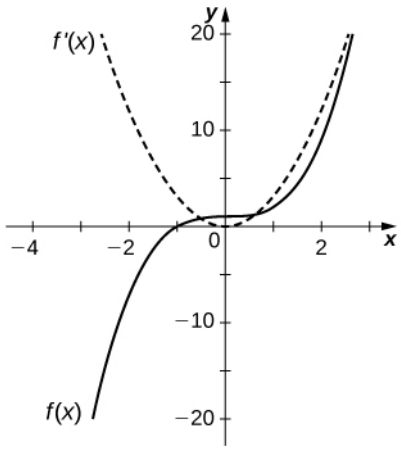
85. $f'(x) = 6x + 2$



87. $f'(x) = -\frac{1}{(2x)^{3/2}}$



89. $f'(x) = 3x^2$



91. a. Average rate at which customers spent on concessions in thousands per customer. b. Rate (in thousands per customer) at which x customers spent money on concessions in thousands per customer.

93. a. Average grade received on the test with an average study time between two values. b. Rate (in percentage points per hour) at which the grade on the test increased or decreased for a given average study time of x hours.

95. a. Average change of atmospheric pressure between two different altitudes. b. Rate (torr per foot) at which atmospheric pressure is increasing or decreasing at x feet.

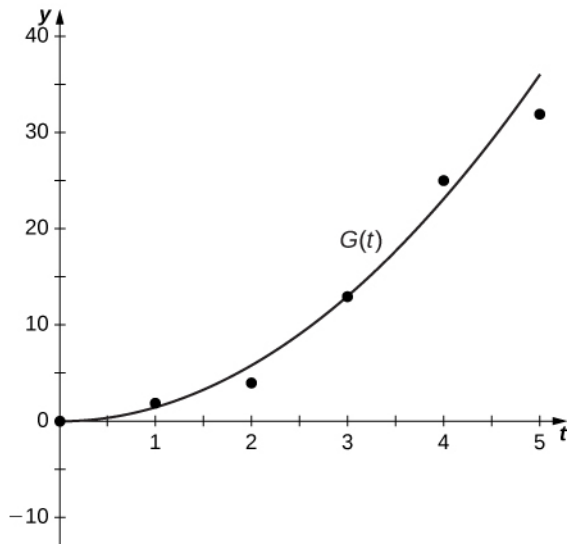
97. a. The rate (in degrees per foot) at which temperature is increasing or decreasing for a given height x. b. The rate of change of temperature as altitude changes at 1000 feet is -0.1 degrees per foot.

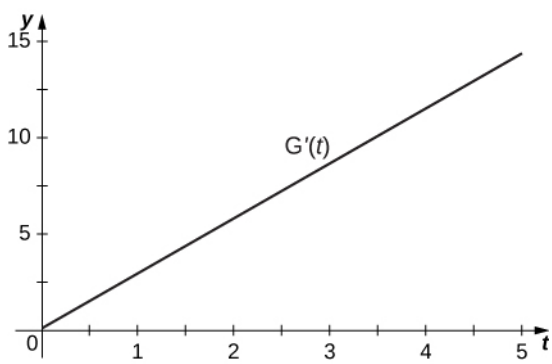
99. a. The rate at which the number of people who have come down with the flu is changing t weeks after the initial outbreak. b. The rate is increasing sharply up to the third week, at which point it slows down and then becomes constant.

101.

Time (seconds)	$h'(t)$ (m/s)
0	2
1	2
2	5.5
3	10.5
4	9.5
5	7

103. $G'(t) = 2.858t + 0.0857$





105. $H''(t) = 0, G''(t) = 2.858$ and $f''(t) = 1.222t + 5.912$ represent the acceleration of the rocket, with units of meters per second squared $\left(\text{m/s}^2\right)$.

Section 3.3 Exercises

107. $f'(x) = 15x^2 - 1$

109. $f'(x) = 32x^3 + 18x$

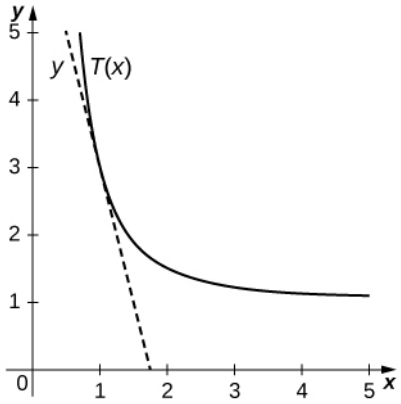
111. $f'(x) = 270x^4 + \frac{39}{(x+1)^2}$

113. $f'(x) = \frac{-5}{x^2}$

115. $f'(x) = \frac{4x^4+2x^2-2x}{x^4}$

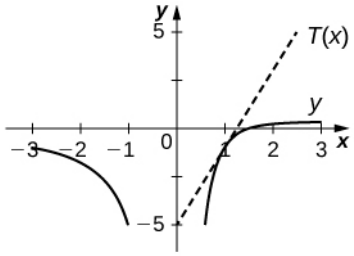
117. $f'(x) = \frac{-x^2-18x+64}{\left(x^2-7x+1\right)^2}$

119.



$T(x) = -4x + 7$

121.



$T(x) = 4x - 5$

123. $h'(x) = 3x^2 f(x) + x^3 f'(x)$

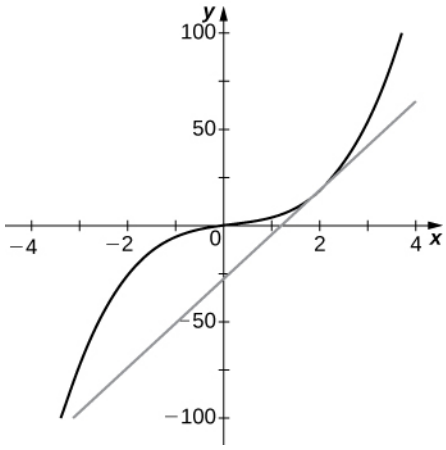
125. $h'(x) = \frac{3f'(x)(g(x)+2)-3f(x)g'(x)}{(g(x)+2)^2}$

127. $\frac{16}{9}$

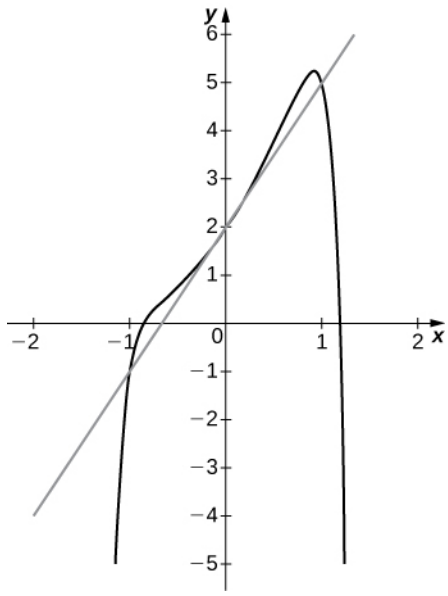
129. Undefined

131. a. 2, b. does not exist, c. 2.5

133. a. 23, b. $y = 23x - 28$



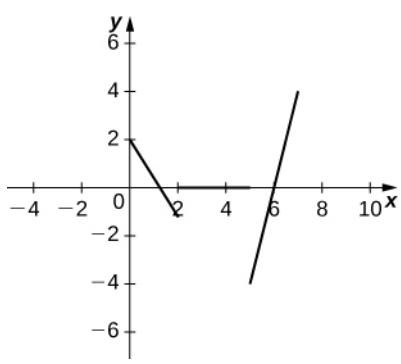
135. a. 3, b. $y = 3x + 2$



137. $y = -7x - 3$
139. $y = -5x + 7$
141. $y = -\frac{3}{2}x + \frac{15}{2}$
143. $y = -3x^2 + 9x - 1$
145. $\frac{12}{121}$ or 0.0992 ft/s
147. a. $\frac{-2t^4-2t^3+200t+50}{\left(t^3+50\right)^2}$ b. -0.02395 mg/L-hr, −0.01344 mg/L-hr, −0.003566 mg/L-hr, −0.001579 mg/L-hr c. The rate at which the concentration of drug in the bloodstream decreases is slowing to 0 as time increases.
149. a. $F'(d) = \frac{-2Gm_1m_2}{d^3}$ b. -1.33×10^{-7} N/m

Section 3.4 Exercises

151. a. $v(t) = 6t^2 - 30t + 36, a(t) = 12t - 30$; b. speeds up $(2, 2.5) \cup \left(3, \infty\right)$, slows down $(0, 2) \cup (2.5, 3)$
153. a. 464 ft/s^2 b. -32 ft/s^2
155. a. 5 ft/s b. 9 ft/s
157. a. 84 ft/s, −84 ft/s b. 84 ft/s c. $\frac{25}{8}$ s d. -32 ft/s^2 in both cases e. $\frac{1}{8}\left(25 + \sqrt{965}\right)$ s f. $-4\sqrt{965}$ ft/s
159. a. Velocity is positive on $(0, 1.5) \cup (6, 7)$, negative on $(1.5, 2) \cup (5, 6)$, and zero on $(2, 5)$. b.



c. Acceleration is positive on $(5, 7)$, negative on $(0, 2)$, and zero on $(2, 5)$. d. The object is speeding up on $(6, 7) \cup (1.5, 2)$ and slowing down on $(0, 1.5) \cup (5, 6)$.
161. a. $R(x) = 10x - 0.001x^2$ b. $R'(x) = 10 - 0.002x$ c. $6 per item, $0 per item
163. a. $C'(x) = 65$ b. $R(x) = 143x - 0.03x^2$, $R'(x) = 143 - 0.06x$ c. $83, -97$. At a production level of 1000 cordless drills, revenue is increasing at a rate of $83 per drill; at a production level of 4000 cordless drills, revenue is decreasing at a rate of $97 per drill. d. $P(x) = -0.03x^2 + 78x - 75000$, $P'(x) = -0.06x + 78$ e. $18, -162$. At a

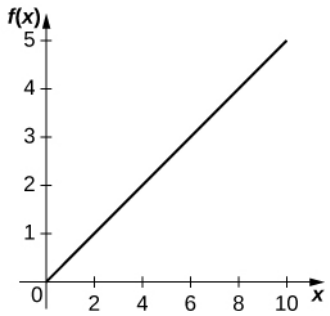
production level of 1000 cordless drills, profit is increasing at a rate of \$18 per drill; at a production level of 4000 cordless drills, profit is decreasing at a rate of \$162 per drill.

165. a. $N'(t) = 3000\left(\frac{-4t^2+400}{\left(t^2+100\right)^2}\right)$ b. $120, 0, -14.4, -9.6$ c. The bacteria population increases from time 0 to 10 hours; afterwards, the bacteria population decreases. d. $0, -6, 0.384, 0.432$. The rate at which the bacteria is increasing is decreasing during the first 10 hours. Afterwards, the bacteria population is decreasing at a decreasing rate.

167. a. $P(t) = 0.03983 + 0.4280$ b. $P'(t) = 0.03983$. The population is increasing. c. $P''(t) = 0$. The rate at which the population is increasing is constant.

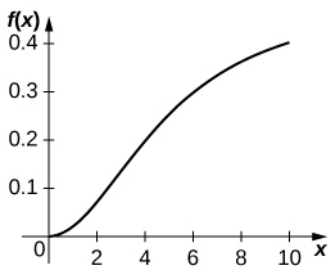
169. a. $p(t) = -0.6071x^2 + 0.4357x - 0.3571$ b. $p'(t) = -1.214x + 0.4357$. This is the velocity of the sensor. c. $p''(t) = -1.214$. This is the acceleration of the sensor; it is a constant acceleration downward.

171. a.



b. $f'(x) = a$. The more increase in prey, the more growth for predators. c. $f''(x) = 0$. As the amount of prey increases, the rate at which the predator population growth increases is constant. d. This equation assumes that if there is more prey, the predator is able to increase consumption linearly. This assumption is unphysical because we would expect there to be some saturation point at which there is too much prey for the predator to consume adequately.

173. a.



b. $f'(x) = \frac{2axn^2}{\left(n^2+x^2\right)^2}$. When the amount of prey increases, the predator growth increases. c. $f''(x) = \frac{2an^2\left(n^2-3x^2\right)}{\left(n^2+x^2\right)^3}$. When the amount of prey is extremely small, the rate at which predator growth is increasing is increasing, but when the amount of prey reaches above a certain threshold, the rate at which predator growth is increasing begins to decrease. d. At lower levels of prey, the prey is more easily able to avoid detection by the predator, so fewer prey individuals are consumed, resulting in less predator growth.

Section 3.5 Exercises

175. $\frac{dy}{dx} = 2x - \sec x \tan x$

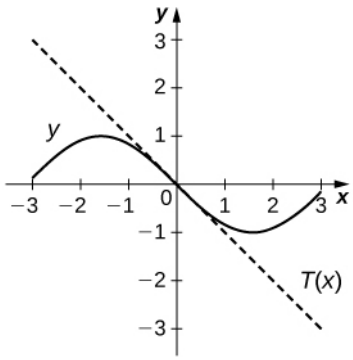
177. $\frac{dy}{dx} = 2x \cot x - x^2 \csc^2 x$

179. $\frac{dy}{dx} = \frac{x \sec x \tan x - \sec x}{x^2}$

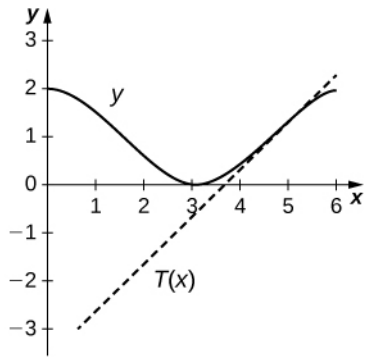
181. $\frac{dy}{dx} = (1 - \sin x)(1 - \sin x) - \cos x (x + \cos x)$

183. $\frac{dy}{dx} = \frac{2\csc^2 x}{(1+\cot x)^2}$

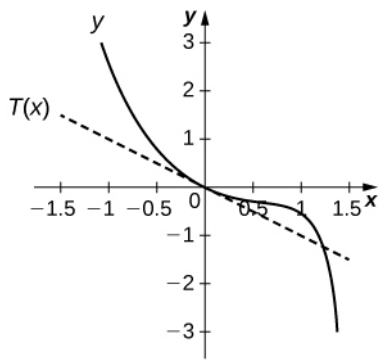
185. $y = -x$



187. $y = x + \frac{2-3\pi}{2}$



189. $y = -x$



191. $3\cos x - x\sin x$

193. $\frac{1}{2}\sin x$

195. $2\csc x\left(\csc^2 x + \cot^2 x\right)$

197. $\frac{(2n+1)\pi}{4}$, where n is an integer

199. $\left(\frac{\pi}{4},\ 1\right),\ \left(\frac{3\pi}{4},\ -1\right),\ \left(\frac{5\pi}{4},\ 1\right),\ \left(\frac{7\pi}{4},\ -1\right)$

201. $a = 0, b = 3$

203. $y' = 5\cos(x)$, increasing on $\left(0, \frac{\pi}{2}\right)$, $\left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$, and $\left(\frac{7\pi}{2}, 12\right)$

209. $3\sin x$

211. $5\cos x$

213. $720x^7 - 5\tan(x)\sec^3(x) - \tan^3(x)\sec(x)$

Section 3.6 Exercises

215. $18u^2 \cdot 7 = 18(7x-4)^2 \cdot 7$

217. $-\sin u \cdot \frac{-1}{8} = -\sin\left(\frac{-x}{8}\right) \cdot \frac{-1}{8}$

219. $\frac{8x-24}{2\sqrt{4u+3}} = \frac{4x-12}{\sqrt{4x^2-24x+3}}$

221. a. $u = 3x^2 + 1$; b. $18x\left(3x^2+1\right)^2$

223. a. $f(u) = u^7, u = \frac{x}{7} + \frac{7}{x}$; b. $7\left(\frac{x}{7} + \frac{7}{x}\right)^6 \cdot \left(\frac{1}{7} - \frac{7}{x^2}\right)$

225. a. $f(u) = \csc u, u = \pi x + 1$; b. $-\pi \csc(\pi x + 1) \cdot \cot(\pi x + 1)$

227. a. $f(u) = -6u^{-3}, u = \sin x$, b. $18(\sin)^{-4} x \cdot \cos x$

229. $\frac{4}{(5-2x)^3}$

231. $6\left(2x^3 - x^2 + 6x + 1\right)^2\left(3x^2 - x + 3\right)$

233. $-3(\tan x + \sin x)^{-4} \cdot \left(\sec^2 x + \cos x\right)$

235. $-7\cos(\cos 7x) \cdot \sin 7x$

237. $-12\cot^2(4x+1) \cdot \csc^2(4x+1)$

239. $10\frac{3}{4}$

241. $y = \frac{-1}{2}x$

243. $x = \pm\sqrt{6}$

245. 10

247. $-\frac{1}{8}$

249. -4

251. -12

253. a. $-\frac{200}{343}$ m/s, b. $\frac{600}{2401}$ m/s^2, c. The train is slowing down since velocity and acceleration have opposite signs.

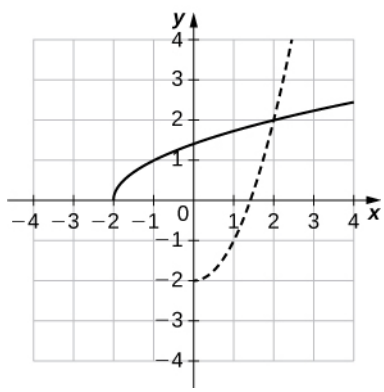
255. a. $C'(x) = 0.0003x^2 - 0.04x + 3$ b. $\frac{dC}{dt} = 100 \cdot \left(0.0003x^2 - 0.04x + 3\right)$ c. Approximately $90,300 per week

257. a. $\frac{dS}{dt} = -\frac{8\pi r^2}{(t+1)^3}$ b. The volume is decreasing at a rate of $-\frac{\pi}{36}$ ft^3/min.

259. ~2.3 ft/hr

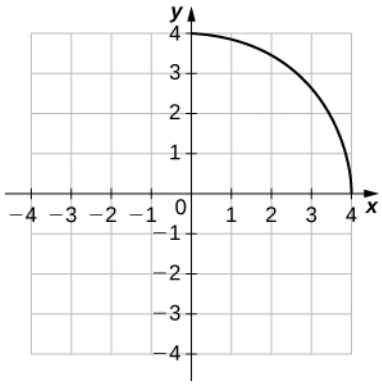
Section 3.7 Exercises

261. a.



b. $\left(f^{-1}\right)'(1) \sim 2$

263. a.



b. $y' = \frac{-x}{\sqrt{16-x^2}}$
$y'(1) = \frac{-1}{\sqrt{16-1}} = \frac{-1}{\sqrt{15}}$

265. a. 6, b. $x = f^{-1}(y) = \left(\frac{y+3}{2}\right)^{1/3}$, c. $\frac{1}{6}$

267. a. 1, b. $x = f^{-1}(y) = \sin^{-1} y$, c. 1

269. $\frac{1}{5}$

271. $\frac{1}{3}$

273. 1

275. a. 4, b. $y = 4x$

277. a. $-\frac{1}{13}$, b. $y = -\frac{1}{13}x + \frac{18}{13}$

279. $\frac{2x}{\sqrt{1-x^4}}$

281. $\frac{-1}{\sqrt{1-x^2}}$

283. $\frac{3\left(1+\tan^{-1} x\right)^2}{1+x^2}$

285. $\frac{-1}{\left(1+x^2\right)\left(\tan^{-1} x\right)^2}$

287. $\frac{x}{\left(5-x^2\right)\sqrt{4-x^2}}$

289. -1

291. $\frac{1}{2}$

293. $\frac{1}{10}$

295. a. $v(t) = \frac{1}{1+t^2}$ b. $a(t) = \frac{-2t}{\left(1+t^2\right)^2}$ c. $(a)\, 0.2, 0.06, 0.03; (b) -0.16, -0.028, -0.0088$ d. The hockey puck is decelerating/slowing down at 2, 4, and 6 seconds.

297. -0.0168 radians per foot

299. a. $\frac{d\theta}{dx} = \frac{10}{100+x^2} - \frac{40}{1600+x^2}$ b. $\frac{18}{325}, \frac{9}{340}, \frac{42}{4745}, 0$ c. As a person moves farther away from the screen, the viewing angle is increasing, which implies that as he or she moves farther away, his or her screen vision is widening. d. $-\frac{54}{12905}, -\frac{3}{500}, -\frac{198}{29945}, -\frac{9}{1360}$ e. As the person moves beyond 20 feet from the screen, the viewing angle is decreasing. The optimal distance the person should stand for maximizing the viewing angle is 20 feet.

Section 3.8 Exercises

301. $\frac{dy}{dx} = \frac{-2x}{y}$

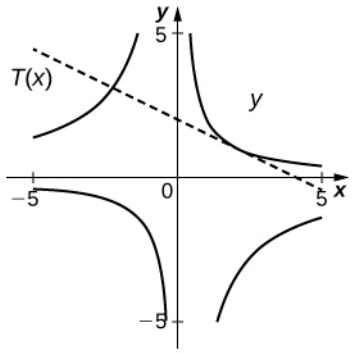
303. $\frac{dy}{dx} = \frac{x}{3y} - \frac{y}{2x}$

305. $\frac{dy}{dx} = \frac{y - \frac{y}{2\sqrt{x+4}}}{\sqrt{x+4}-x}$

307. $\frac{dy}{dx} = \frac{y^2\cos(xy)}{2y-\sin(xy)-xy\cos(xy)}$

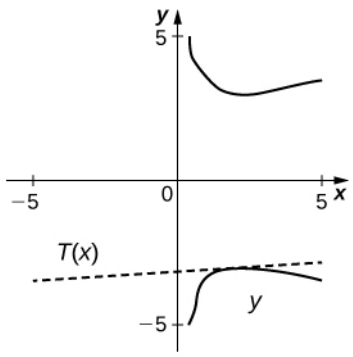
309. $\frac{dy}{dx} = \frac{-3x^2y-y^3}{x^3+3xy^2}$

311.



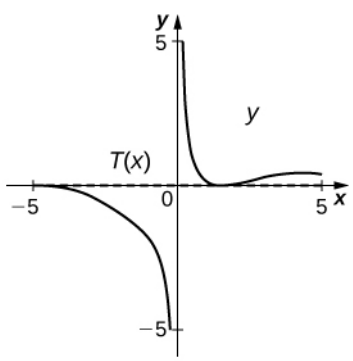
$y = \frac{-1}{2}x + 2$

313.



$y = \frac{1}{\pi+12}x - \frac{3\pi+38}{\pi+12}$

315.



$y = 0$

317. a. $y = -x + 2$ b. $(3, -1)$

319. a. $\left(\pm\sqrt{7}, 0\right)$ b. -2 c. They are parallel since the slope is the same at both intercepts.

321. $y = -x + 1$

323. a. -0.5926 b. When \$81 is spent on labor and \$16 is spent on capital, the amount spent on capital is decreasing by \$0.5926 per \$1 spent on labor.

325. -8

327. -2.67

329. $y' = -\frac{1}{\sqrt{1-x^2}}$

Section 3.9 Exercises

331. $2xe^x + x^2e^x$

333. $e^{x^3 \ln x}\left(3x^2 \ln x + x^2\right)$

335. $\frac{4}{\left(e^x+e^{-x}\right)^2}$

337. $2^{4x+2} \cdot \ln 2 + 8x$

339. $\pi x^{\pi-1} \cdot \pi^x + x^\pi \cdot \pi^x \ln \pi$

341. $\frac{5}{2(5x-7)}$

343. $\frac{\tan x}{\ln 10}$

345. $2^x \cdot \ln 2 \cdot \log_3 7^{x^2-4} + 2^x \cdot \frac{2x \ln 7}{\ln 3}$

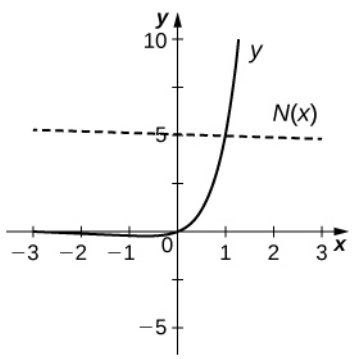
347. $(\sin 2x)^{4x}\,[4 \cdot \ln(\sin 2x) + 8x \cdot \cot 2x]$

349. $x^{\log_2 x} \cdot \frac{2 \ln x}{x \ln 2}$

351. $x^{\cot x} \cdot \left[-\csc^2 x \cdot \ln x + \frac{\cot x}{x}\right]$

353. $x^{-1/2}\left(x^2+3\right)^{2/3}(3x-4)^4 \cdot \left[\frac{-1}{2x} + \frac{4x}{3\left(x^2+3\right)} + \frac{12}{3x-4}\right]$

355.



$y = \frac{-1}{5+5\ln 5}x + \left(5 + \frac{1}{5+5\ln 5}\right)$

357. a. $x = e \sim 2.718$ b. $y' > 0$ on $(0, e)$; $y' < 0$ on (e, ∞)

359. a. $P = 500{,}000(1.05)^t$ individuals b. $P'(t) = 24395 \cdot (1.05)^t$ individuals per year c. 39,737 individuals per year

361. a. At the beginning of 1960 there were 5.3 thousand cases of the disease in New York City. At the beginning of 1964 there were approximately 723 cases of the disease in the United States. b. At the beginning of 1960 the number of cases of the disease was decreasing at rate of -4.611 thousand per year; at the beginning of 1963, the number of cases of the disease was decreasing at a rate of -0.2808 thousand per year.

363. $p = 35741(1.045)^t$

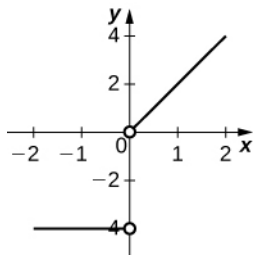
365.

Years since 1790	P''
0	69.25
10	107.5
20	167.0
30	259.4
40	402.8
50	625.5

Years since 1790	P''
60	971.4
70	1508.5

Review Exercises

367. False.
369. False
371. $\frac{1}{2\sqrt{x+4}}$
373. $9x^2 + \frac{8}{x^3}$
375. $e^{\sin x}\cos x$
377. $x\sec^2(x) + 2x\cos(x) + \tan(x) - x^2\sin(x)$
379. $\frac{1}{4}\left(\frac{x}{\sqrt{1-x^2}} + \sin^{-1}(x)\right)$
381. $\cos x \cdot (\ln x + 1) - x\ln(x)\sin x$
383. $4^x(\ln 4)^2 + 2\sin x + 4x\cos x - x^2\sin x$
385. $T = (2+e)x - 2$
387.



389. $w'(3) = -\frac{2.9\pi}{6}$. At 3 a.m. the tide is decreasing at a rate of 1.518 ft/hr.
391. -7.5. The wind speed is decreasing at a rate of 7.5 mph/hr

Chapter 4

Checkpoint

4.1 $\frac{1}{72\pi}$ cm/sec, or approximately 0.0044 cm/sec
4.2 500 ft/sec
4.3 $\frac{1}{10}$ rad/sec
4.4 -0.61 ft/sec
4.5 $L(x) = 2 + \frac{1}{12}(x-8)$; 2.00833
4.6 $L(x) = -x + \frac{\pi}{2}$
4.7 $L(x) = 1 + 4x$
4.8 $dy = 2xe^{x^2}\,dx$
4.9 $dy = 1.6$, $\Delta y = 1.64$
4.10 The volume measurement is accurate to within 21.6 cm^3.
4.11 7.6%
4.12 $x = -\frac{2}{3}$, $x = 1$
4.13 The absolute maximum is 3 and it occurs at $x = 4$. The absolute minimum is -1 and it occurs at $x = 2$.
4.14 $c = 2$
4.15 $\frac{5}{2\sqrt{2}}$ sec
4.16 f has a local minimum at -2 and a local maximum at 3.
4.17 f has no local extrema because f' does not change sign at $x = 1$.

4.18 f is concave up over the interval $\left(-\infty, \frac{1}{2}\right)$ and concave down over the interval $\left(\frac{1}{2}, \infty\right)$

4.19 f has a local maximum at -2 and a local minimum at 3.

4.20 Both limits are 3. The line $y = 3$ is a horizontal asymptote.

4.21 Let $\varepsilon > 0$. Let $N = \frac{1}{\sqrt{\varepsilon}}$. Therefore, for all $x > N$, we have

$$\left|3 - \frac{1}{x^2} - 3\right| = \frac{1}{x^2} < \frac{1}{N^2} = \varepsilon$$

Therefore, $\lim\limits_{x \to \infty} \left(3 - 1/x^2\right) = 3.$

4.22 Let $M > 0$. Let $N = \sqrt{\frac{M}{3}}$. Then, for all $x > N$, we have

$$3x^2 > 3N^2 = 3\left(\sqrt{\frac{M}{3}}\right)^2{}^2 = \frac{3M}{3} = M$$

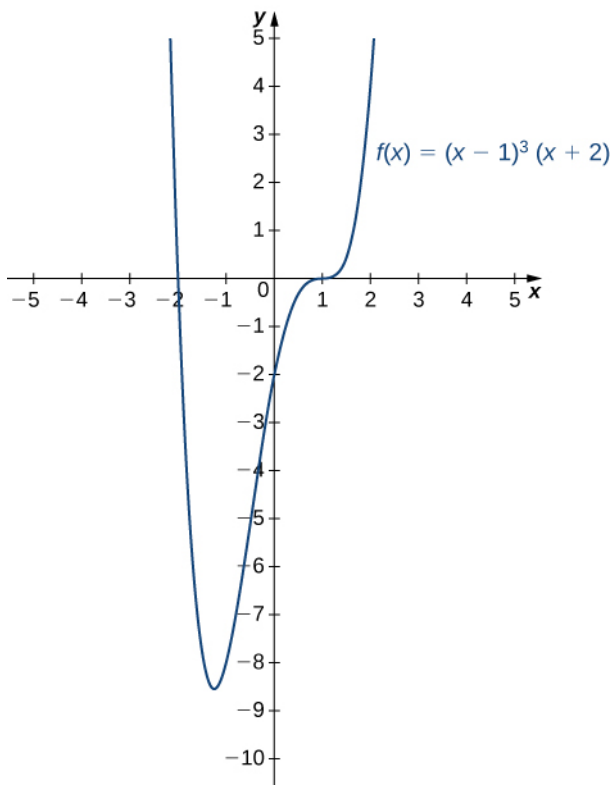
4.23 $-\infty$

4.24 $\frac{3}{5}$

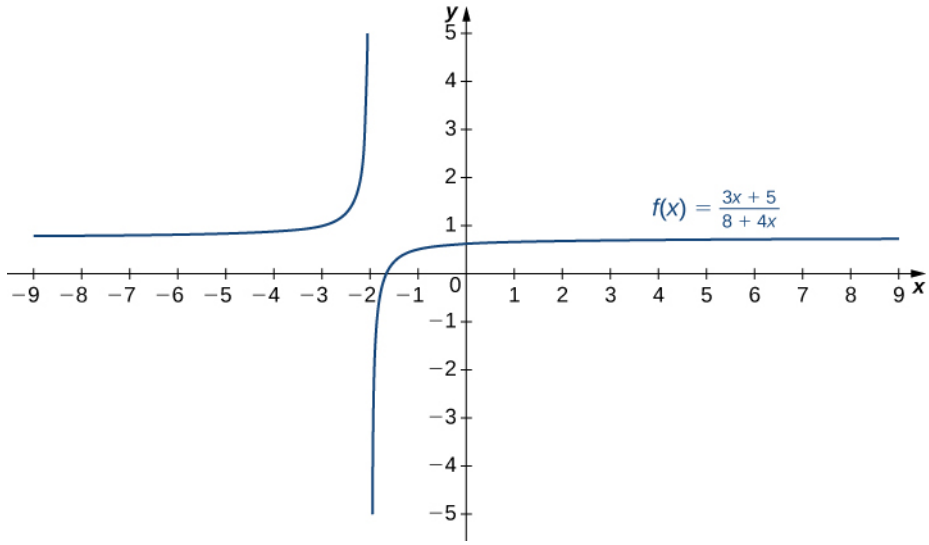
4.25 $\pm\sqrt{3}$

4.26 $\lim\limits_{x \to \infty} f(x) = \frac{3}{5}, \quad \lim\limits_{x \to -\infty} f(x) = -2$

4.27



4.28



4.29 $y = \frac{3}{2}x$

4.30 The function f has a cusp at $(0, 5)$ $\lim_{x \to 0^-} f'(x) = \infty$, $\lim_{x \to 0^+} f'(x) = -\infty$. For end behavior, $\lim_{x \to \pm\infty} f(x) = -\infty$.

4.31 The maximum area is 5000 ft^2.

4.32 $V(x) = x(20 - 2x)(30 - 2x)$. The domain is $[0, 10]$.

4.33 $T(x) = \frac{x}{6} + \frac{\sqrt{(15-x)^2+1}}{2.5}$

4.34 The company should charge \$75 per car per day.

4.35 $A(x) = 4x\sqrt{1 - x^2}$. The domain of consideration is $[0, 1]$.

4.36 $c(x) = \frac{259.2}{x} + 0.2x^2$ dollars

4.37 1

4.38 0

4.39 $\lim_{x \to 0^+} \cos x = 1$. Therefore, we cannot apply L'Hôpital's rule. The limit of the quotient is ∞

4.40 1

4.41 0

4.42 e

4.43 1

4.44 The function 2^x grows faster than x^{100}.

4.45 $x_1 \approx 0.33333333, x_2 \approx 0.347222222$

4.46 $x_1 = 2, x_2 = 1.75$

4.47 $x_1 \approx -1.842105263, x_2 \approx -1.772826920$

4.48 $x_1 = 6, x_2 = 8, x_3 = \frac{26}{3}, x_4 = \frac{80}{9}, x_5 = \frac{242}{27}; x^* = 9$

4.49 $-\cos x + C$

4.50 $\frac{d}{dx}(x \sin x + \cos x + C) = \sin x + x \cos x - \sin x = x \cos x$

4.51 $x^4 - \frac{5}{3}x^3 + \frac{1}{2}x^2 - 7x + C$

4.52 $y = -\frac{3}{x} + 5$

4.53 $2.93 \text{ sec}, \ 64.5 \text{ ft}$

Section 4.1 Exercises

1. 8
3. $\pm\frac{13}{\sqrt{10}}$
5. $2\sqrt{3}$ ft/sec
7. The distance is decreasing at 390 mi/h.
9. The distance between them shrinks at a rate of $\frac{1320}{13} \approx 101.5$ mph.
11. $\frac{9}{2}$ ft/sec

13. It grows at a rate $\frac{4}{9}$ ft/sec

15. The distance is increasing at $\frac{\left(135\sqrt{26}\right)}{26}$ ft/sec

17. $-\frac{5}{6}$ m/sec

19. 240π m^2/sec

21. $\frac{1}{2\sqrt{\pi}}$ cm

23. The area is increasing at a rate $\frac{\left(3\sqrt{3}\right)}{8}$ $\text{ft}^2/\text{sec}.$

25. The depth of the water decreases at $\frac{128}{125\pi}$ ft/min.

27. The volume is decreasing at a rate of $\frac{(25\pi)}{16}\text{ft}^3/\text{min}.$

29. The water flows out at rate $\frac{(2\pi)}{5}$ $\text{m}^3/\text{min}.$

31. $\frac{3}{2}$ m/sec

33. $\frac{25}{19\pi}$ ft/min

35. $\frac{2}{45\pi}$ ft/min

37. The angle decreases at $\frac{400}{1681}$ $\text{rad/sec}.$

39. $100\pi\text{mi/min}$

41. The angle is changing at a rate of $\frac{11}{25}$ $\text{rad/sec}.$

43. The distance is increasing at a rate of 62.50 ft/sec.

45. The distance is decreasing at a rate of 11.99 ft/sec.

Section 4.2 Exercises

47. $f'(a) = 0$

49. The linear approximation exact when $y = f(x)$ is linear or constant.

51. $L(x) = \frac{1}{2} - \frac{1}{4}(x-2)$

53. $L(x) = 1$

55. $L(x) = 0$

57. 0.02

59. 1.9996875

61. 0.001593

63. 1; error, ~0.00005

65. 0.97; error, ~0.0006

67. $3 - \frac{1}{600}$; error, ~4.632×10^{-7}

69. $dy = (\cos x - x\sin x)\,dx$

71. $dy = \left(\frac{x^2-2x-2}{(x-1)^2}\right)dx$

73. $dy = -\frac{1}{(x+1)^2}dx, -\frac{1}{16}$

75. $dy = \frac{9x^2+12x-2}{2(x+1)^{3/2}}dx, -0.1$

77. $dy = \left(3x^2 + 2 - \frac{1}{x^2}\right)dx, 0.2$

79. $12x\,dx$

81. $4\pi r^2\,dr$

83. $-1.2\pi\ \text{cm}^3$

85. $-100\ \text{ft}^3$

Section 4.3 Exercises

91. Answers may vary

93. Answers will vary

95. No; answers will vary

97. Since the absolute maximum is the function (output) value rather than the *x* value, the answer is no; answers will vary

99. When $a = 0$

101. Absolute minimum at 3; Absolute maximum at −2.2; local minima at −2, 1; local maxima at −1, 2

103. Absolute minima at −2, 2; absolute maxima at −2.5, 2.5; minima at −2, 0, and 2 (Note: −2 and 2 are both local and absolute minima); local maxima at −1, 1
105. Answers may vary.
107. Answers may vary.
109. $x = 1$
111. None
113. $x = 0; x = \pm 2$
115. None
117. $x = -1, 1$
119. Absolute maximum: $x = 4$, $y = \frac{33}{2}$; absolute minimum: $x = 1$, $y = 3$
121. Absolute minimum: $x = \frac{1}{2}$, $y = 4$
123. Absolute maximum: $x = 2\pi$, $y = 2\pi$; absolute minimum: $x = 0$, $y = 0$
125. Absolute maximum: $x = -3$ and $y = 6$; ; absolute minimum: $-1 \le x \le 1$, $y = 2$
127. Absolute maximum: $x = \frac{\pi}{4}$, $y = \sqrt{2}$; absolute minimum: $x = \frac{5\pi}{4}$, $y = -\sqrt{2}$
129. Absolute minimum: $x = -2$, $y = 1$
131. Absolute minimum: $x = -3$, $y = -135$; local maximum: $x = 0$, $y = 0$; local minimum: $x = 1$, $y = -7$
133. Local maximum: $x = 1 - 2\sqrt{2}$, $y = 3 - 4\sqrt{2}$; local minimum: $x = 1 + 2\sqrt{2}$, $y = 3 + 4\sqrt{2}$
135. Absolute maximum: $x = \frac{\sqrt{2}}{2}$, $y = \frac{3}{2}$; absolute minimum: $x = -\frac{\sqrt{2}}{2}$, $y = -\frac{3}{2}$
137. Local maximum: $x = -2$, $y = 59$; local minimum: $x = 1$, $y = -130$
139. Absolute maximum: $x = 0$, $y = 1$; absolute minimum: $x = -2, 2$, $y = 0$
141. $h = \frac{9245}{49}$ m, $t = \frac{300}{49}$ s
143. The absolute minimum was in 1848, when no gold was produced.
145. Absolute minima: $x = 0$, $x = 2$, $y = 1$; local maximum at $x = 1$, $y = 2$
147. No maxima/minima if a is odd, minimum at $x = 1$ if a is even

Section 4.4 Exercises

149. One example is $f(x) = |x| + 3,\ -2 \le x \le 2$
151. Yes, but the Mean Value Theorem still does not apply
153. Any closed interval in $\left(-\infty, 0\right)$ or $\left(0, \infty\right)$
155. Any closed interval in $\left(-\infty, -2\right)$ or $\left(2, \infty\right)$
157. 2 points
159. 5 points
161. $c = \frac{2\sqrt{3}}{3}$
163. $c = \frac{1}{2}, 1, \frac{3}{2}$
165. $c = 1$
167. Not differentiable
169. Not differentiable
171. Yes
173. The Mean Value Theorem does not apply since the function is discontinuous at $x = \frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}$.
175. Yes
177. The Mean Value Theorem does not apply; discontinuous at $x = 0$.
179. Yes
181. The Mean Value Theorem does not apply; not differentiable at $x = 0$.
183. $b = \pm 2\sqrt{c}$
185. $c = \pm \frac{1}{\pi} \cos^{-1}\left(\frac{\sqrt{\pi}}{2}\right)$, $c = \pm 0.1533$
187. The Mean Value Theorem does not apply.
189. $\frac{1}{2\sqrt{c+1}} - \frac{2}{c^3} = \frac{521}{2880}$; $c = 3.133, 5.867$
191. Yes
193. Let $s_1(t)$ represent the distance of Car 1 from Stoplight 1 at time t. Let $s_2(t)$ represent the distance of Car 2 from Stoplight 1 at time t. Then $s_1(0) = 0 = s_2(0)$. Let $t = m$ represent the time that the cars arrive at Stoplight 2. Then $s_1(m) = d = s_2(m)$ where d is the distance between the stoplights.

Define a new function $f(t) = s_1(t) - s_2(t)$. Then $f(0) = 0 = f(m)$. By the Mean Value Theorem, there is a time $0 < c < m$ such that

$$f'\left(c\right) = \frac{f(m) - f(0)}{m - 0} = \frac{0}{m} = 0.$$

At time c,

$$f'\left(c\right) = 0$$
$$s_1'\left(c\right) - s_2'\left(c\right) = 0$$
$$s_1'\left(c\right) = s_2'\left(c\right)$$

So, at time c, Car 1 and Car 2 are traveling at the same speed.

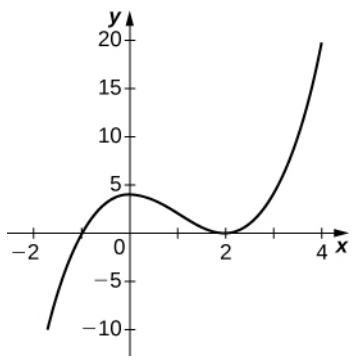
Section 4.5 Exercises

195. It is not a local maximum/minimum because f' does not change sign
197. No
199. False; for example, $y = \sqrt{x}$.
201. Increasing for $-2 < x < -1$ and $x > 2$; decreasing for $x < -2$ and $-1 < x < 2$
203. Decreasing for $x < 1$, increasing for $x > 1$
205. Decreasing for $-2 < x < -1$ and $1 < x < 2$; increasing for $-1 < x < 1$ and $x < -2$ and $x > 2$
207. a. Increasing over $-2 < x < -1, 0 < x < 1, x > 2$, decreasing over $x < -2, -1 < x < 0, 1 < x < 2$; b. maxima at $x = -1$ and $x = 1$, minima at $x = -2$ and $x = 0$ and $x = 2$
209. a. Increasing over $x > 0$, decreasing over $x < 0$; b. Minimum at $x = 0$
211. Concave up on all x, no inflection points
213. Concave up on all x, no inflection points
215. Concave up for $x < 0$ and $x > 1$, concave down for $0 < x < 1$, inflection points at $x = 0$ and $x = 1$
217. Answers will vary
219. Answers will vary
221. a. Increasing over $-\frac{\pi}{2} < x < \frac{\pi}{2}$, decreasing over $x < -\frac{\pi}{2}, x > \frac{\pi}{2}$ b. Local maximum at $x = \frac{\pi}{2}$; local minimum at $x = -\frac{\pi}{2}$
223. a. Concave up for $x > \frac{4}{3}$, concave down for $x < \frac{4}{3}$ b. Inflection point at $x = \frac{4}{3}$
225. a. Increasing over $x < 0$ and $x > 4$, decreasing over $0 < x < 4$ b. Maximum at $x = 0$, minimum at $x = 4$ c. Concave up for $x > 2$, concave down for $x < 2$ d. Infection point at $x = 2$
227. a. Increasing over $x < 0$ and $x > \frac{60}{11}$, decreasing over $0 < x < \frac{60}{11}$ b. Minimum at $x = \frac{60}{11}$, local maximum at x = 0 c. Concave down for $x < \frac{54}{11}$, concave up for $x > \frac{54}{11}$ d. Inflection point at $x = \frac{54}{11}$
229. a. Increasing over $x > -\frac{1}{2}$, decreasing over $x < -\frac{1}{2}$ b. Minimum at $x = -\frac{1}{2}$ c. Concave up for all x d. No inflection points
231. a. Increases over $-\frac{1}{4} < x < \frac{3}{4}$, decreases over $x > \frac{3}{4}$ and $x < -\frac{1}{4}$ b. Minimum at $x = -\frac{1}{4}$, maximum at $x = \frac{3}{4}$ c. Concave up for $-\frac{3}{4} < x < \frac{1}{4}$, concave down for $x < -\frac{3}{4}$ and $x > \frac{1}{4}$ d. Inflection points at $x = -\frac{3}{4}, x = \frac{1}{4}$
233. a. Increasing for all x b. No local minimum or maximum c. Concave up for $x > 0$, concave down for $x < 0$ d. Inflection point at $x = 0$
235. a. Increasing for all x where defined b. No local minima or maxima c. Concave up for $x < 1$; concave down for $x > 1$ d. No inflection points in domain
237. a. Increasing over $-\frac{\pi}{4} < x < \frac{3\pi}{4}$, decreasing over $x > \frac{3\pi}{4}, x < -\frac{\pi}{4}$ b. Minimum at $x = -\frac{\pi}{4}$, maximum at $x = \frac{3\pi}{4}$ c. Concave up for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, concave down for $x < -\frac{\pi}{2}, x > \frac{\pi}{2}$ d. Infection points at $x = \pm\frac{\pi}{2}$
239. a. Increasing over $x > 4$, decreasing over $0 < x < 4$ b. Minimum at $x = 4$ c. Concave up for $0 < x < 8\sqrt[3]{2}$, concave down for $x > 8\sqrt[3]{2}$ d. Inflection point at $x = 8\sqrt[3]{2}$
241. $f > 0, f' > 0, f'' < 0$
243. $f > 0, f' < 0, f'' > 0$
245. $f > 0, f' > 0, f'' > 0$
247. True, by the Mean Value Theorem
249. True, examine derivative

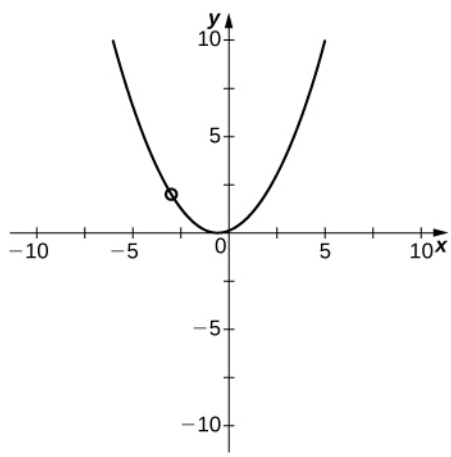
Section 4.6 Exercises

251. $x = 1$

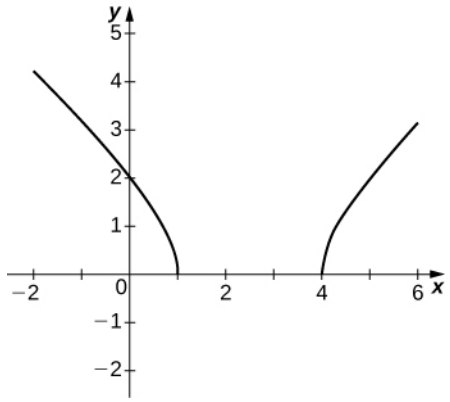
253. $x = -1, x = 2$
255. $x = 0$
257. Yes, there is a vertical asymptote
259. Yes, there is vertical asymptote
261. 0
263. ∞
265. $-\frac{1}{7}$
267. -2
269. -4
271. Horizontal: none, vertical: $x = 0$
273. Horizontal: none, vertical: $x = \pm 2$
275. Horizontal: none, vertical: none
277. Horizontal: $y = 0$, vertical: $x = \pm 1$
279. Horizontal: $y = 0$, vertical: $x = 0$ and $x = -1$
281. Horizontal: $y = 1$, vertical: $x = 1$
283. Horizontal: none, vertical: none
285. Answers will vary, for example: $y = \frac{2x}{x-1}$
287. Answers will vary, for example: $y = \frac{4x}{x+1}$
289. $y = 0$
291. ∞
293. $y = 3$
295.



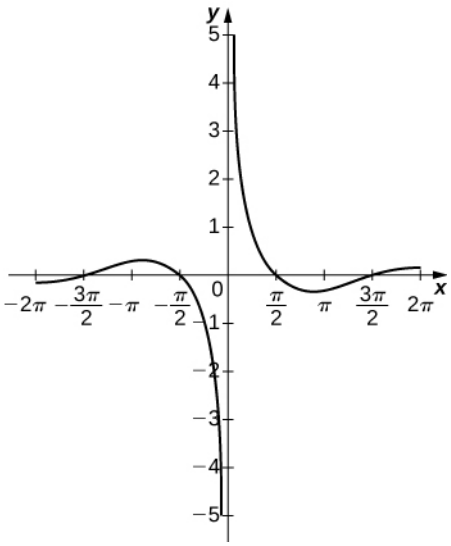
297.



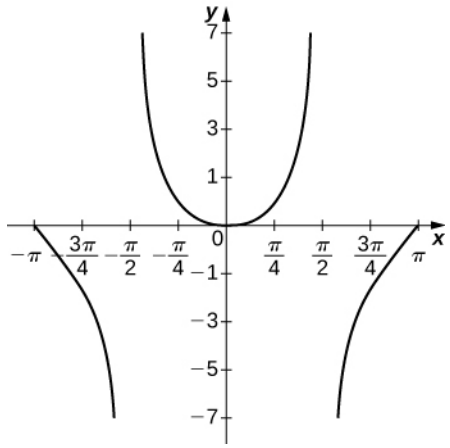
299.


y
5
4
3
2
1
−2
0
2
4
6
x
−1
−2

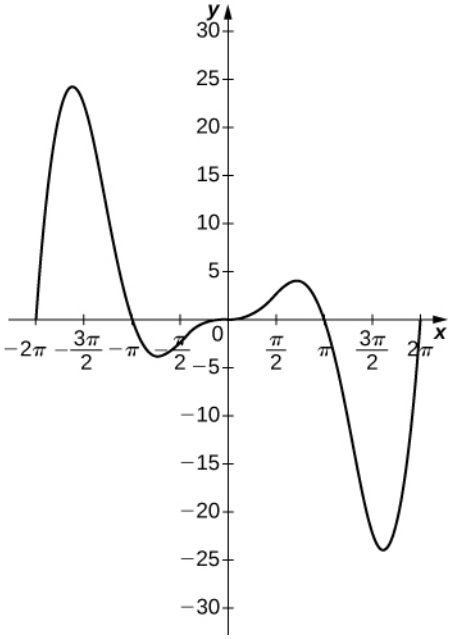
301.


y
5
4
3
2
1
0
x
−2π
−3π/2
−π
−π/2
π/2
π
3π/2
2π
−1
−2
−3
−4
−5

303.


y
7
5
3
1
0
x
−π
−3π/4
−π/2
−π/4
π/4
π/2
3π/4
π
−1
−3
−5
−7

305.



307. $P(0) \neq 0$ and $Q(0) = 0$

309. $\lim_{x \to 1^-} f(x) = -\infty$ and $\lim_{x \to 1^-} g(x) = \infty$

Section 4.7 Exercises

311. The critical points can be the minima, maxima, or neither.

313. False; $y = x^2$ has a minimum only

315. $h = \frac{62}{3}$ in.

317. 1

319. 100 ft by 100 ft

321. 40 ft by 40 ft

323. 19.73 ft.

325. 84 bpm

327. $T(\theta) = \frac{40\theta}{3v} + \frac{40\cos\theta}{v}$

329. $v = \sqrt{\frac{b}{a}}$

331. approximately 34 mph

333. 4

335. 0

337. Maximal: $x = 5, y = 5$; minimal: $x = 0, y = 10$ and $y = 0, x = 10$

339. Maximal: $x = 1, y = 9$; minimal: none

341. $\frac{4\pi}{3\sqrt{3}}$

343. 6

345. $r = 2, h = 4$

347. $(2, 1)$

349. $(0.8351, 0.6974)$

351. $A = 20\left(\frac{20}{\pi+4}\right) - 2\left(\frac{20}{\pi+4}\right)^2 - \frac{1}{2}\pi\left(\frac{20}{\pi+4}\right)^2$

353. $C(x) = 5x^2 + \frac{32}{x}$

355. $P(x) = (50 - x)(800 + 25x - 50)$

Section 4.8 Exercises

357. ∞

359. $\frac{1}{2a}$

361. $\frac{1}{na^{n-1}}$
363. Cannot apply directly; use logarithms
365. Cannot apply directly; rewrite as $\lim_{x \to 0} x^3$
367. 6
369. -2
371. -1
373. n
375. $-\frac{1}{2}$
377. $\frac{1}{2}$
379. 1
381. $\frac{1}{6}$
383. 1
385. 0
387. 0
389. -1
391. ∞
393. 0
395. $\frac{1}{e}$
397. 0
399. 1
401. 0
403. $\tan(1)$
405. 2

Section 4.9 Exercises

407. $F(x_n) = x_n - \frac{{x_n}^3+2x_n+1}{3{x_n}^2+2}$
409. $F(x_n) = x_n - \frac{e^{x_n}}{e^{x_n}}$
411. $|c| > 0.5$ fails, $|c| \leq 0.5$ works
413. $c = \frac{1}{f'(x_n)}$
415. a. $x_1 = \frac{12}{25}, x_2 = \frac{312}{625}$; b. $x_1 = -4, x_2 = -40$
417. a. $x_1 = 1.291, x_2 = 0.8801$; b. $x_1 = 0.7071, x_2 = 1.189$
419. a. $x_1 = -\frac{26}{25}, x_2 = -\frac{1224}{625}$; b. $x_1 = 4, x_2 = 18$
421. a. $x_1 = \frac{6}{10}, x_2 = \frac{6}{10}$; b. $x_1 = 2, x_2 = 2$
423. 3.1623 or -3.1623
425. $0, -1$ or 1
427. 0
429. 0.5188 or -1.2906
431. 0
433. 4.493
435. $0.159, 3.146$
437. We need f to be twice continuously differentiable.
439. $x = 0$
441. $x = -1$
443. $x = 5.619$
445. $x = -1.326$
447. There is no solution to the equation.
449. It enters a cycle.
451. 0
453. -0.3513
455. Newton: 11 iterations, secant: 16 iterations
457. Newton: three iterations, secant: six iterations
459. Newton: five iterations, secant: eight iterations
461. $E = 4.071$

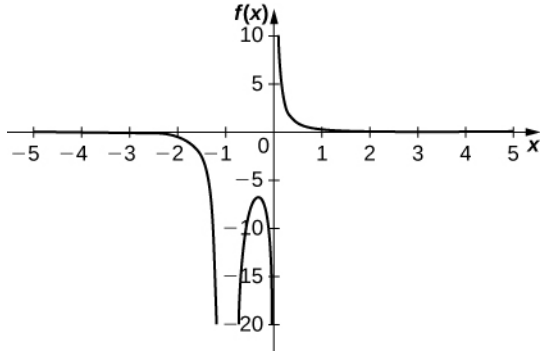
463. 4.394%

Section 4.10 Exercises

465. $F'(x) = 15x^2 + 4x + 3$
467. $F'(x) = 2xe^x + x^2e^x$
469. $F'(x) = e^x$
471. $F(x) = e^x - x^3 - \cos(x) + C$
473. $F(x) = \frac{x^2}{2} - x - 2\cos(2x) + C$
475. $F(x) = \frac{1}{2}x^2 + 4x^3 + C$
477. $F(x) = \frac{2}{5}\left(\sqrt{x}\right)^5 + C$
479. $F(x) = \frac{3}{2}x^{2/3} + C$
481. $F(x) = x + \tan(x) + C$
483. $F(x) = \frac{1}{3}\sin^3(x) + C$
485. $F(x) = -\frac{1}{2}\cot(x) - \frac{1}{x} + C$
487. $F(x) = -\sec x - 4\csc x + C$
489. $F(x) = -\frac{1}{8}e^{-4x} - \cos x + C$
491. $-\cos x + C$
493. $3x - \frac{2}{x} + C$
495. $\frac{8}{3}x^{3/2} + \frac{4}{5}x^{5/4} + C$
497. $14x - \frac{2}{x} - \frac{1}{2x^2} + C$
499. $f(x) = -\frac{1}{2x^2} + \frac{3}{2}$
501. $f(x) = \sin x + \tan x + 1$
503. $f(x) = -\frac{1}{6}x^3 - \frac{2}{x} + \frac{13}{6}$
505. Answers may vary; one possible answer is $f(x) = e^{-x}$
507. Answers may vary; one possible answer is $f(x) = -\sin x$
509. 5.867 sec
511. 7.333 sec
513. 13.75 ft/sec^2
515. $F(x) = \frac{1}{3}x^3 + 2x$
517. $F(x) = x^2 - \cos x + 1$
519. $F(x) = -\frac{1}{(x+1)} + 1$
521. True
523. False

Review Exercises

525. True, by Mean Value Theorem
527. True
529. Increasing: $(-2, 0) \cup \left(4, \infty\right)$, decreasing: $\left(-\infty, -2\right) \cup (0, 4)$
531. $L(x) = \frac{17}{16} + \frac{1}{2}(1 + 4\pi)\left(x - \frac{1}{4}\right)$
533. Critical point: $x = \frac{3\pi}{4}$, absolute minimum: $x = 0$, absolute maximum: $x = \pi$
535. Increasing: $(-1, 0) \cup \left(3, \infty\right)$, decreasing: $\left(-\infty, -1\right) \cup (0, 3)$, concave up: $\left(-\infty, \frac{1}{3}\left(2 - \sqrt{13}\right)\right) \cup \left(\frac{1}{3}\left(2 + \sqrt{13}\right), \infty\right)$, concave down: $\left(\frac{1}{3}\left(2 - \sqrt{13}\right), \frac{1}{3}\left(2 + \sqrt{13}\right)\right)$
537. Increasing: $\left(\frac{1}{4}, \infty\right)$, decreasing: $\left(0, \frac{1}{4}\right)$, concave up: $\left(0, \infty\right)$, concave down: nowhere
539. 3
541. $-\frac{1}{\pi}$
543. $x_1 = -1, x_2 = -1$
545. $F(x) = \frac{2x^{3/2}}{3} + \frac{1}{x} + C$
547.



Inflection points: none; critical points: $x = -\frac{1}{3}$; zeros: none; vertical asymptotes: $x = -1$, $x = 0$; horizontal asymptote: $y = 0$

549. The height is decreasing at a rate of 0.125 m/sec

551. $x = \sqrt{ab}$ feet

Chapter 5

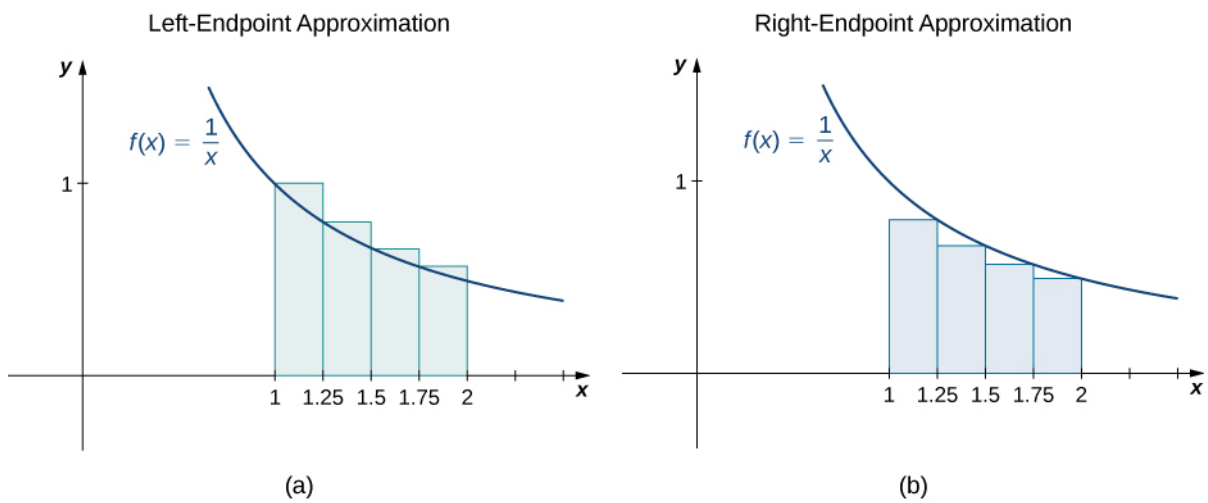
Checkpoint

5.1 $\sum_{i=3}^{6} 2^i = 2^3 + 2^4 + 2^5 + 2^6 = 120$

5.2 15,550

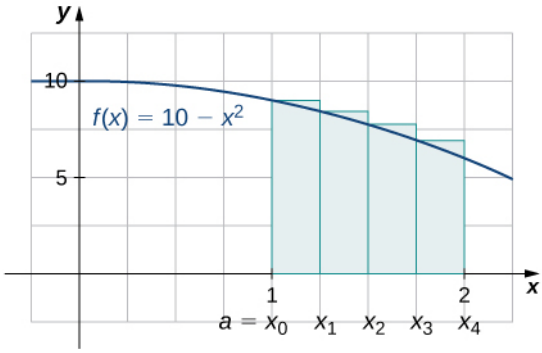
5.3 440

5.4 The left-endpoint approximation is 0.7595. The right-endpoint approximation is 0.6345.



5.5 a. Upper sum $= 8.0313$.

b.

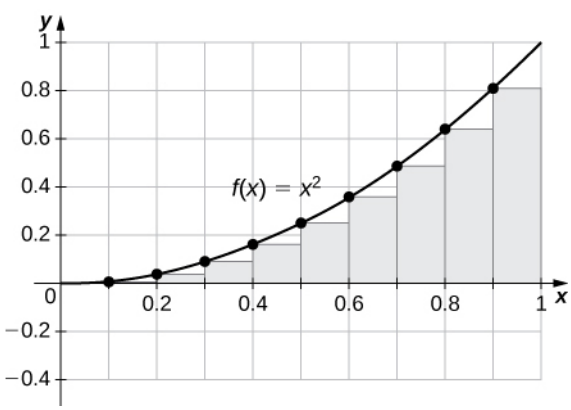


5.6 $A \approx 1.125$
5.7 6
5.8 18 square units
5.9 6
5.10 18
5.11 $6\int_1^3 x^3 dx - 4\int_1^3 x^2 dx + 2\int_1^3 x dx - \int_1^3 3dx$
5.12 −7
5.13 3
5.14 Average value $= 1.5; c = 3$
5.15 $c = \sqrt{3}$
5.16 $g'(r) = \sqrt{r^2+4}$
5.17 $F'(x) = 3x^2 \cos x^3$
5.18 $F'(x) = 2x\cos x^2 - \cos x$
5.19 $\frac{7}{24}$
5.20 Kathy still wins, but by a much larger margin: James skates 24 ft in 3 sec, but Kathy skates 29.3634 ft in 3 sec.
5.21 $-\frac{10}{3}$
5.22 Net displacement: $\frac{e^2-9}{2} \approx -0.8055$ m; total distance traveled: $4\ln 4 - 7.5 + \frac{e^2}{2} \approx 1.740$ m
5.23 17.5 mi
5.24 $\frac{64}{5}$
5.25 $\int 3x^2\left(x^3-3\right)^2 dx = \frac{1}{3}\left(x^3-3\right)^3 + C$
5.26 $\frac{\left(x^3+5\right)^{10}}{30} + C$
5.27 $-\frac{1}{\sin t} + C$
5.28 $-\frac{\cos^4 t}{4} + C$
5.29 $\frac{91}{3}$
5.30 $\frac{2}{3\pi} \approx 0.2122$
5.31 $\int x^2 e^{-2x^3} dx = -\frac{1}{6}e^{-2x^3} + C$
5.32 $\int e^x(3e^x-2)^2 dx = \frac{1}{9}(3e^x-2)^3$
5.33 $\int 2x^3 e^{x^4} dx = \frac{1}{2}e^{x^4}$
5.34 $\frac{1}{2}\int_0^4 e^u du = \frac{1}{2}\left(e^4-1\right)$
5.35 $Q(t) = \frac{2^t}{\ln 2} + 8.557$. There are 20,099 bacteria in the dish after 3 hours.
5.36 There are 116 flies.
5.37 $\int_1^2 \frac{1}{x^3} e^{4x^{-2}} dx = \frac{1}{8}\left[e^4 - e\right]$
5.38 $\ln|x+2| + C$

5.39 $\frac{x}{\ln 3}(\ln x - 1) + C$
5.40 $\frac{1}{4}\sin^{-1}(4x) + C$
5.41 $\sin^{-1}\left(\frac{x}{3}\right) + C$
5.42 $\frac{1}{10}\tan^{-1}\left(\frac{2x}{5}\right) + C$
5.43 $\frac{1}{4}\tan^{-1}\left(\frac{x}{4}\right) + C$
5.44 $\frac{\pi}{8}$

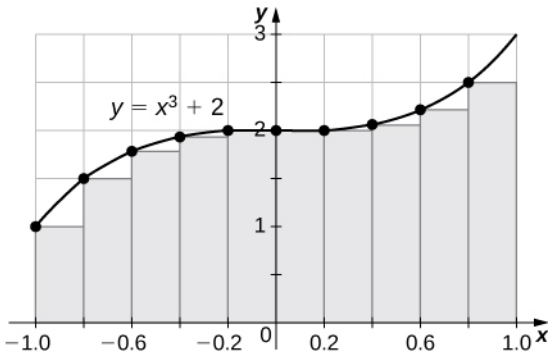
Section 5.1 Exercises

1. a. They are equal; both represent the sum of the first 10 whole numbers. b. They are equal; both represent the sum of the first 10 whole numbers. c. They are equal by substituting $j = i - 1$. d. They are equal; the first sum factors the terms of the second.
3. $385 - 30 = 355$
5. $15 - (-12) = 27$
7. $5(15) + 4(-12) = 27$
9. $\sum_{j=1}^{50} j^2 - 2\sum_{j=1}^{50} j = \frac{(50)(51)(101)}{6} - \frac{2(50)(51)}{2} = 40, \quad 375$
11. $4\sum_{k=1}^{25} k^2 - 100\sum_{k=1}^{25} k = \frac{4(25)(26)(51)}{6} - 50(25)(26) = -10, \quad 400$
13. $R_4 = -0.25$
15. $R_6 = 0.372$
17. $L_4 = 2.20$
19. $L_8 = 0.6875$
21. $L_6 = 9.000 = R_6$. The graph of f is a triangle with area 9.
23. $L_6 = 13.12899 = R_6$. They are equal.
25. $L_{10} = \frac{4}{10}\sum_{i=1}^{10}\sqrt{4 - \left(-2 + 4\frac{(i-1)}{10}\right)^2}$
27. $R_{100} = \frac{e-1}{100}\sum_{i=1}^{100}\ln\left(1 + (e-1)\frac{i}{100}\right)$
29.



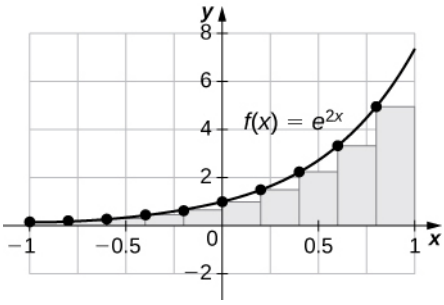
$R_{100} = 0.33835$, $L_{100} = 0.32835$. The plot shows that the left Riemann sum is an underestimate because the function is increasing. Similarly, the right Riemann sum is an overestimate. The area lies between the left and right Riemann sums. Ten rectangles are shown for visual clarity. This behavior persists for more rectangles.

31.



$L_{100} = 3.980, R_{100} = 4.020, R_{100} > L_{100}$. The left endpoint sum is an underestimate because the function is increasing. Similarly, a right endpoint approximation is an overestimate. The area lies between the left and right endpoint estimates.

33.



$L_{100} = 3.555, R_{100} = 3.670$ The plot shows that the left Riemann sum is an underestimate because the function is increasing. Ten rectangles are shown for visual clarity. This behavior persists for more rectangles.

35. The sum represents the cumulative rainfall in January 2009.

37. The total mileage is $7 \times \sum_{i=1}^{25} \left(1 + \frac{(i-1)}{10}\right) = 7 \times 25 + \frac{7}{10} \times 12 \times 25 = 385 \text{ mi}.$

39. Add the numbers to get 8.1-in. net increase.

41. 309,389,957

43. $L_8 = 3 + 2 + 1 + 2 + 3 + 4 + 5 + 4 = 24$

45. $L_8 = 3 + 5 + 7 + 6 + 8 + 6 + 5 + 4 = 44$

47. $L_{10} \approx 1.7604, L_{30} \approx 1.7625, L_{50} \approx 1.76265$

49. $R_1 = -1, L_1 = 1, R_{10} = -0.1, L_{10} = 0.1, L_{100} = 0.01$, and $R_{100} = -0.01$. By symmetry of the graph, the exact area is zero.

51. $R_1 = 0, L_1 = 0, R_{10} = 2.4499, L_{10} = 2.4499, R_{100} = 2.1365, L_{100} = 2.1365$

53. If $[c, d]$ is a subinterval of $[a, b]$ under one of the left-endpoint sum rectangles, then the area of the rectangle contributing to the left-endpoint estimate is $f(c)(d-c)$. But, $f(c) \le f(x)$ for $c \le x \le d$, so the area under the graph of *f* between *c* and *d* is $f(c)(d-c)$ plus the area below the graph of *f* but above the horizontal line segment at height $f(c)$, which is positive. As this is true for each left-endpoint sum interval, it follows that the left Riemann sum is less than or equal to the area below the graph of *f* on $[a, b]$.

55. $L_N = \frac{b-a}{N} \sum_{i=1}^{N} f\left(a + (b-a)\frac{i-1}{N}\right) = \frac{b-a}{N} \sum_{i=0}^{N-1} f\left(a + (b-a)\frac{i}{N}\right)$ and $R_N = \frac{b-a}{N} \sum_{i=1}^{N} f\left(a + (b-a)\frac{i}{N}\right)$.
The left sum has a term corresponding to $i = 0$ and the right sum has a term corresponding to $i = N$. In $R_N - L_N$, any term corresponding to $i = 1, 2, \ldots, N-1$ occurs once with a plus sign and once with a minus sign, so each such term cancels and one is left with
$R_N - L_N = \frac{b-a}{N}\left(f(a + (b-a))\frac{N}{N}\right) - \left(f(a) + (b-a)\frac{0}{N}\right) = \frac{b-a}{N}(f(b) - f(a))$.

57. Graph 1: a. $L(A) = 0, B(A) = 20$; b. $U(A) = 20$. Graph 2: a. $L(A) = 9$; b. $B(A) = 11, U(A) = 20$. Graph 3: a. $L(A) = 11.0$; b. $B(A) = 4.5, U(A) = 15.5$.

59. Let *A* be the area of the unit circle. The circle encloses *n* congruent triangles each of area $\frac{\sin\left(\frac{2\pi}{n}\right)}{2}$, so $\frac{n}{2}\sin\left(\frac{2\pi}{n}\right) \le A$. Similarly, the circle is contained inside *n* congruent triangles each of area

$\frac{BH}{2} = \frac{1}{2}\left(\cos\left(\frac{\pi}{n}\right)+\sin\left(\frac{\pi}{n}\right)\tan\left(\frac{\pi}{n}\right)\right)^2 \sin\left(\frac{2\pi}{n}\right)$, so $A \leq \frac{n}{2}\sin\left(\frac{2\pi}{n}\right)\left(\cos\left(\frac{\pi}{n}\right)+\sin\left(\frac{\pi}{n}\right)\tan\left(\frac{\pi}{n}\right)\right)^2$. As $n \to \infty$, $\frac{n}{2}\sin\left(\frac{2\pi}{n}\right) = \frac{\sin\left(\frac{2\pi}{n}\right)}{\left(\frac{2}{n}\right)} A \to \pi$, so we conclude $\pi \leq A$. Also, as $n \to \infty$, $\cos\left(\frac{\pi}{n}\right)+\sin\left(\frac{\pi}{n}\right)\tan\left(\frac{\pi}{n}\right) \to 1$, so we also have $A \leq \pi$. By the squeeze theorem for limits, we conclude that $A = \pi$.

Section 5.2 Exercises

61. $\int_0^2 \left(5x^2 - 3x^3\right) dx$

63. $\int_0^1 \cos^2(2\pi x)\, dx$

65. $\int_0^1 x dx$

67. $\int_3^6 x dx$

69. $\int_1^2 x \log\left(x^2\right) dx$

71. $1 + 2 \cdot 2 + 3 \cdot 3 = 14$

73. $1 - 4 + 9 = 6$

75. $1 - 2\pi + 9 = 10 - 2\pi$

77. The integral is the area of the triangle, $\frac{1}{2}$

79. The integral is the area of the triangle, 9.

81. The integral is the area $\frac{1}{2}\pi r^2 = 2\pi$.

83. The integral is the area of the "big" triangle less the "missing" triangle, $9 - \frac{1}{2}$.

85. $L = 2 + 0 + 10 + 5 + 4 = 21$, $R = 0 + 10 + 10 + 2 + 0 = 22$, $\frac{L+R}{2} = 21.5$

87. $L = 0 + 4 + 0 + 4 + 2 = 10$, $R = 4 + 0 + 2 + 4 + 0 = 10$, $\frac{L+R}{2} = 10$

89. $\int_2^4 f(x)\, dx + \int_2^4 g(x)\, dx = 8 - 3 = 5$

91. $\int_2^4 f(x)\, dx - \int_2^4 g(x)\, dx = 8 + 3 = 11$

93. $4\int_2^4 f(x)\, dx - 3\int_2^4 g(x)\, dx = 32 + 9 = 41$

95. The integrand is odd; the integral is zero.

97. The integrand is antisymmetric with respect to $x = 3$. The integral is zero.

99. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$

101. $\int_0^1 \left(1 - 6x + 12x^2 - 8x^3\right) dx = \left(x - 3x^2 + 4x^3 - 2x^4\right) = \left(1 - 3 + 4 - 2\right)\left(0 - 0 + 0 - 0\right) = 0$

103. $7 - \frac{5}{4} = \frac{23}{4}$

105. The integrand is negative over $[-2, 3]$.

107. $x \leq x^2$ over $[1, 2]$, so $\sqrt{1+x} \leq \sqrt{1+x^2}$ over $[1, 2]$.

109. $\cos(t) \geq \frac{\sqrt{2}}{2}$. Multiply by the length of the interval to get the inequality.

111. $f_{\text{ave}} = 0; c = 0$

113. $\frac{3}{2}$ when $c = \pm\frac{3}{2}$

115. $f_{\text{ave}} = 0; c = \frac{\pi}{2}, \frac{3\pi}{2}$

117. $L_{100} = 1.294$, $R_{100} = 1.301$; the exact average is between these values.

119. $L_{100} \times \left(\frac{1}{2}\right) = 0.5178$, $R_{100} \times \left(\frac{1}{2}\right) = 0.5294$

121. $L_1 = 0$, $L_{10} \times \left(\frac{1}{2}\right) = 8.743493$, $L_{100} \times \left(\frac{1}{2}\right) = 12.861728$. The exact answer ≈ 26.799, so L_{100} is not accurate.

123. $L_1 \times \left(\frac{1}{\pi}\right) = 1.352$, $L_{10} \times \left(\frac{1}{\pi}\right) = -0.1837$, $L_{100} \times \left(\frac{1}{\pi}\right) = -0.2956$. The exact answer ≈ -0.303, so L_{100} is not accurate to first decimal.

125. Use $\tan^2\theta + 1 = \sec^2\theta$. Then, $B - A = \int_{-\pi/4}^{\pi/4} 1 dx = \frac{\pi}{2}$.

127. $\int_0^{2\pi} \cos^2 t dt = \pi$, so divide by the length 2π of the interval. $\cos^2 t$ has period π, so yes, it is true.

129. The integral is maximized when one uses the largest interval on which p is nonnegative. Thus, $A = \frac{-b-\sqrt{b^2-4ac}}{2a}$ and $B = \frac{-b+\sqrt{b^2-4ac}}{2a}$.

131. If $f(t_0) > g(t_0)$ for some $t_0 \in [a, b]$, then since $f - g$ is continuous, there is an interval containing t_0 such that $f(t) > g(t)$ over the interval $[c, d]$, and then $\int_d^d f(t)\,dt > \int_c^d g(t)\,dt$ over this interval.

133. The integral of f over an interval is the same as the integral of the average of f over that interval. Thus,
$$\int_a^b f(t)\,dt = \int_{a_0}^{a_1} f(t)\,dt + \int_{a_1}^{a_2} f(t)\,dt + \cdots + \int_{a_{N+1}}^{a_N} f(t)\,dt = \int_{a_0}^{a_1} 1dt + \int_{a_1}^{a_2} 1dt + \cdots + \int_{a_{N+1}}^{a_N} 1dt$$
$$= (a_1 - a_0) + (a_2 - a_1) + \cdots + \left(a_N - a_{N-1}\right) = a_N - a_0 = b - a.$$
Dividing through by $b - a$ gives the desired identity.

135. $\int_0^N f(t)\,dt = \sum_{i=1}^{N} \int_{i-1}^{i} f(t)\,dt = \sum_{i=1}^{N} i^2 = \frac{N(N+1)(2N+1)}{6}$

137. $L_{10} = 1.815, R_{10} = 1.515, \frac{L_{10}+R_{10}}{2} = 1.665$, so the estimate is accurate to two decimal places.

139. The average is $1/2$, which is equal to the integral in this case.

141. a. The graph is antisymmetric with respect to $t = \frac{1}{2}$ over $[0, 1]$, so the average value is zero. b. For any value of a, the graph between $[a, a + 1]$ is a shift of the graph over $[0, 1]$, so the net areas above and below the axis do not change and the average remains zero.

143. Yes, the integral over any interval of length 1 is the same.

Section 5.3 Exercises

145. Yes. It is implied by the Mean Value Theorem for Integrals.

147. $F'(2) = -1$; average value of F' over $[1, 2]$ is $-1/2$.

149. $e^{\cos x}$

151. $\frac{1}{\sqrt{16-x^2}}$

153. $\sqrt{x}\frac{d}{dx}\sqrt{x} = \frac{1}{2}$

155. $-\sqrt{1-\cos^2 x}\frac{d}{dx}\cos x = |\sin x|\sin x$

157. $2x\frac{|x|}{1+x^2}$

159. $\ln(e^{2x})\frac{d}{dx}e^x = 2xe^x$

161. a. f is positive over $(1, 2)$ and $(5, 6)$, negative over $(0, 1)$ and $(3, 4)$, and zero over $(2, 3)$ and $(4, 5)$. b. The maximum value is 2 and the minimum is –3. c. The average value is 0.

163. a. ℓ is positive over $(0, 1)$ and $(3, 6)$, and negative over $(1, 3)$. b. It is increasing over $(0, 1)$ and $(3, 5)$, and it is constant over $(1, 3)$ and $(5, 6)$. c. Its average value is $\frac{1}{3}$.

165. $T_{10} = 49.08, \int_{-4}^{3} \left(x^3 + 6x^2 + x - 5\right)dx = 48$

167. $T_{10} = 260.836, \int_1^9 \left(\sqrt{x} + x^2\right)dx = 260$

169. $T_{10} = 3.058, \int_1^4 \frac{4}{x^2}dx = 3$

171. $F(x) = \frac{x^3}{3} + \frac{3x^2}{2} - 5x, F(3) - F(-2) = -\frac{35}{6}$

173. $F(x) = -\frac{t^5}{5} + \frac{13t^3}{3} - 36t, F(3) - F(2) = \frac{62}{15}$

175. $F(x) = \frac{x^{100}}{100}, F(1) - F(0) = \frac{1}{100}$

177. $F(x) = \frac{x^3}{3} + \frac{1}{x}, F(4) - F\left(\frac{1}{4}\right) = \frac{1125}{64}$

179. $F(x) = \sqrt{x}, F(4) - F(1) = 1$

181. $F(x) = \frac{4}{3}t^{3/4}, F(16) - F(1) = \frac{28}{3}$

183. $F(x) = -\cos x, F\left(\frac{\pi}{2}\right) - F(0) = 1$

185. $F(x) = \sec x, F\left(\frac{\pi}{4}\right) - F(0) = \sqrt{2} - 1$

187. $F(x) = -\cot(x), F\left(\frac{\pi}{2}\right) - F\left(\frac{\pi}{4}\right) = 1$

189. $F(t) = -\frac{1}{t} + \frac{1}{2t^2}, F(-1) - F(-2) = \frac{7}{8}$

191. $F(x) = e^x - e$

193. $F(x) = 0$

195. $\int_{-2}^{-1} \left(t^2 - 2t - 3\right) dt - \int_{-1}^{3} \left(t^2 - 2t - 3\right) dt + \int_{3}^{4} \left(t^2 - 2t - 3\right) dt = \frac{46}{3}$

197. $-\int_{-\pi/2}^{0} \sin t dt + \int_{0}^{\pi/2} \sin t dt = 2$

199. a. The average is 11.21×10^9 since $\cos\left(\frac{\pi t}{6}\right)$ has period 12 and integral 0 over any period. Consumption is equal to the average when $\cos\left(\frac{\pi t}{6}\right) = 0$, when $t = 3$, and when $t = 9$. b. Total consumption is the average rate times duration: $11.21 \times 12 \times 10^9 = 1.35 \times 10^{11}$ c.
$10^9 \left(11.21 - \frac{1}{6}\int_3^9 \cos\left(\frac{\pi t}{6}\right) dt\right) = 10^9 \left(11.21 + \frac{2}{\pi}\right) = 11.84x10^9$

201. If f is not constant, then its average is strictly smaller than the maximum and larger than the minimum, which are attained over $[a, b]$ by the extreme value theorem.

203. a. $d^2\theta = (a\cos\theta + c)^2 + b^2\sin^2\theta = a^2 + c^2\cos^2\theta + 2ac\cos\theta = (a + c\cos\theta)^2$; b. $\overline{d} = \frac{1}{2\pi}\int_0^{2\pi} (a + c\cos\theta)\, d\theta = a$

205. Mean gravitational force = $\frac{GmM}{2\pi}\int_0^{2\pi} \frac{1}{\left(a + 2\sqrt{a^2 - b^2}\cos\theta\right)^2} d\theta.$

Section 5.4 Exercises

207. $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right) dx = \int x^{1/2} dx - \int x^{-1/2} dx = \frac{2}{3}x^{3/2} + C_1 - 2x^{1/2} + C_2 = \frac{2}{3}x^{3/2} - 2x^{1/2} + C$

209. $\int \frac{dx}{2x} = \frac{1}{2}\ln|x| + C$

211. $\int_0^{\pi} \sin x dx - \int_0^{\pi} \cos x dx = -\cos x|_0^{\pi} - (\sin x)|_0^{\pi} = (-(-1) + 1) - (0 - 0) = 2$

213. $P(s) = 4s$, so $\frac{dP}{ds} = 4$ and $\int_2^4 4ds = 8.$

215. $\int_1^2 N ds = N$

217. With p as in the previous exercise, each of the 12 pentagons increases in area from $2p$ to $4p$ units so the net increase in the area of the dodecahedron is $36p$ units.

219. $18s^2 = 6\int_s^{2s} 2x dx$

221. $12\pi R^2 = 8\pi \int_R^{2R} r dr$

223. $d(t) = \int_0^t v(s)\, ds = 4t - t^2$. The total distance is $d(2) = 4$ m.

225. $d(t) = \int_0^t v(s)\, ds$. For $t < 3, d(t) = \int_0^t (6 - 2t)\, dt = 6t - t^2$. For
$t > 3, d(t) = d(3) + \int_3^t (2t - 6) dt = 9 + (t^2 - 6t)\Big|_3^6$. The total distance is $d(6) = 18$ m.

227. $v(t) = 40 - 9.8t$m/sec; $h(t) = 1.5 + 40t - 4.9t^2$ m/s

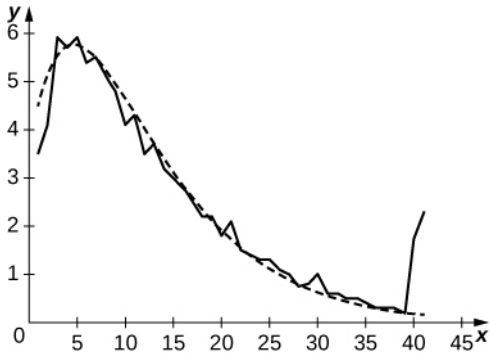
229. The net increase is 1 unit.

231. At $t = 5$, the height of water is $x = \left(\frac{15}{\pi}\right)^{1/3}$ m.. The net change in height from $t = 5$ to $t = 10$ is $\left(\frac{30}{\pi}\right)^{1/3} - \left(\frac{15}{\pi}\right)^{1/3}$ m.

233. The total daily power consumption is estimated as the sum of the hourly power rates, or 911 gW-h.

235. 17 kJ

237. a. 54.3%; b. 27.00%; c. The curve in the following plot is $2.35\,(t + 3)\, e^{-0.15(t+3)}$.

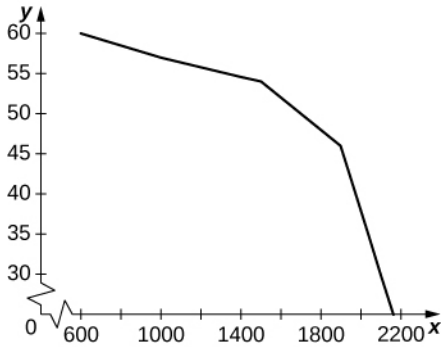


239. In dry conditions, with initial velocity $v_0 = 30$ m/s, $D = 64.3$ and, if $v_0 = 25$, $D = 44.64$. In wet conditions, if $v_0 = 30$, and $D = 180$ and if $v_0 = 25$, $D = 125$.

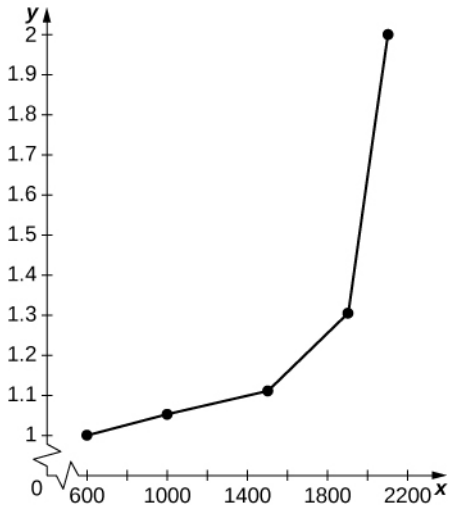
241. 225 cal

243. $E(150) = 28$, $E(300) = 22$, $E(450) = 16$

245. a.



b. Between 600 and 1000 the average decrease in vehicles per hour per lane is −0.0075. Between 1000 and 1500 it is −0.006 per vehicles per hour per lane, and between 1500 and 2100 it is −0.04 vehicles per hour per lane. c.



The graph is nonlinear, with minutes per mile increasing dramatically as vehicles per hour per lane reach 2000.

247. $\frac{1}{37}\int_0^{37} p(t)\,dt = \frac{0.07(37)^3}{4} + \frac{2.42(37)^2}{3} - \frac{25.63(37)}{2} + 521.23 \approx 2037$

249. Average acceleration is $-\frac{0.7}{3}t^3 + \frac{1.44}{2}t^2 + 10.44t\Big|_0^5 = -\frac{0.7}{3}\left(5^3\right) + 10.44\left(5\right) \approx 41$ mph/s

251. $s\left(t\right) = v\left(t\right)dt = \int\left(-\frac{7}{30}t^3 + 0.72t^2 + 10.44t + 23.967\right)dt = -\frac{7}{120}t^4 + \frac{0.72}{3}t^3 + \frac{10.44}{2}t^2 + 23.967t + C = 174.24$ feet

253. $\frac{1}{40}\int_0^{40}(-0.068t+5.14)\,dt = -\frac{0.068(40)}{2}+5.14 = 3.78\text{m/sec}$

Section 5.5 Exercises

255. $u = h(x)$

257. $f(u) = \frac{(u+1)^2}{\sqrt{u}}$

259. $du = 8xdx;\ f(u) = \frac{1}{8\sqrt{u}}$

261. $\frac{1}{5}(x+1)^5 + C$

263. $-\frac{1}{12(2x-3)^6} + C$

265. $\sqrt{x^2+1} + C$

267. $\frac{1}{8}\left(x^2-2x\right)^4 + C$

269. $\sin\theta - \frac{\sin^3\theta}{3} + C$

271. $\frac{(1-x)^{101}}{101} - \frac{(1-x)^{100}}{100} + C$

273. $\int(11x-7)^{-2}\text{dx} = -\frac{1}{22(11x-7)^2} + C$

275. $-\frac{\cos^4\theta}{4} + C$

277. $-\frac{\cos^3(\pi t)}{3\pi} + C$

279. $-\frac{1}{4}\cos^2\left(t^2\right) + C$

281. $-\frac{1}{3(x^3-3)} + C$

283. $-\frac{2\left(y^3-2\right)}{3\sqrt{1-y^3}} + C$

285. $\frac{1}{33}\left(1-\cos^3\theta\right)^{11} + C$

287. $\frac{1}{12}\left(\sin^3\theta - 3\sin^2\theta\right)^4 + C$

289. $L_{50} = -8.5779$. The exact area is $\frac{-81}{8}$

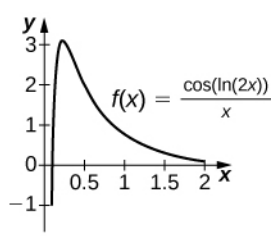
291. $L_{50} = -0.006399$ ... The exact area is 0.

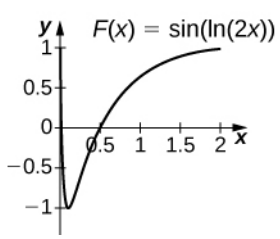
293. $u = 1+x^2, du = 2xdx, \frac{1}{2}\int_1^2 u^{-1/2}du = \sqrt{2}-1$

295. $u = 1+t^3, du = 3t^2dt, \frac{1}{3}\int_1^2 u^{-1/2}du = \frac{2}{3}\left(\sqrt{2}-1\right)$

297. $u = \cos\theta, du = -\sin\theta d\theta, \int_{1/\sqrt{2}}^1 u^{-4}du = \frac{1}{3}\left(2\sqrt{2}-1\right)$

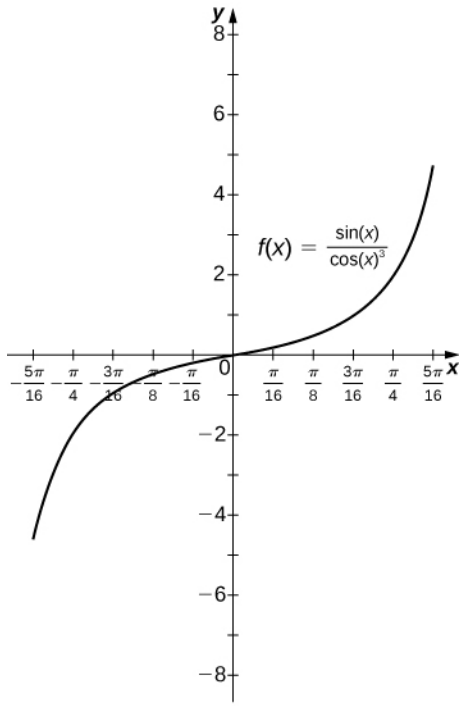
299.

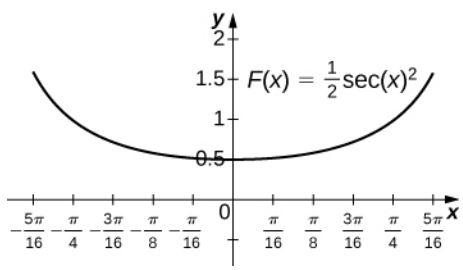




The antiderivative is $y = \sin(\ln(2x))$. Since the antiderivative is not continuous at $x = 0$, one cannot find a value of C that would make $y = \sin(\ln(2x)) - C$ work as a definite integral.

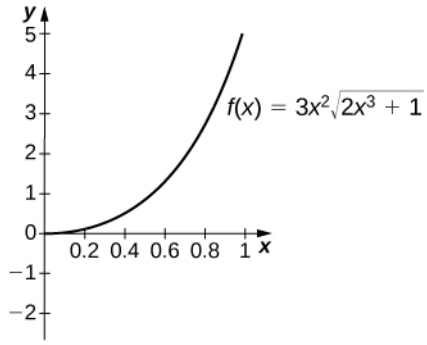
301.

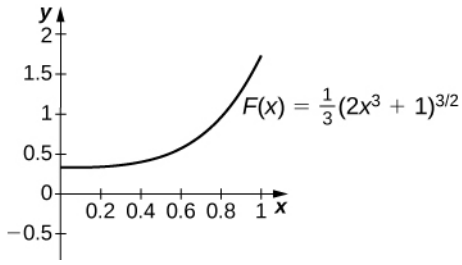




The antiderivative is $y = \frac{1}{2}\sec^2 x$. You should take $C = -2$ so that $F\left(-\frac{\pi}{3}\right) = 0$.

303.





The antiderivative is $y = \frac{1}{3}\left(2x^3 + 1\right)^{3/2}$. One should take $C = -\frac{1}{3}$.

305. No, because the integrand is discontinuous at $x = 1$.

307. $u = \sin\left(t^2\right)$; the integral becomes $\frac{1}{2}\int_0^0 u du$.

309. $u = \left(1 + \left(t - \frac{1}{2}\right)^2\right)$; the integral becomes $-\int_{5/4}^{5/4} \frac{1}{u} du$.

311. $u = 1 - t$; the integral becomes

$$\int_1^{-1} u\cos(\pi(1-u))\,du$$
$$= \int_1^{-1} u\left[\cos\pi\cos\pi u - \sin\pi\sin\pi u\right]du$$
$$= -\int_1^{-1} u\cos\pi u du$$
$$= \int_{-1}^{1} u\cos\pi u du = 0$$

since the integrand is odd.

313. Setting $u = cx$ and $du = cdx$ gets you $\frac{1}{\frac{b}{c}-\frac{a}{c}}\int_{a/c}^{b/c} f(cx)\,dx = \frac{c}{b-a}\int_{u=a}^{u=b} f(u)\frac{du}{c} = \frac{1}{b-a}\int_a^b f(u)\,du.$

315. $\int_0^x g(t)\,dt = \frac{1}{2}\int_{u=1-x^2}^{1} \frac{du}{u^a} = \frac{1}{2(1-a)}u^{1-a}\Big|_{u=1-x^2}^{1} = \frac{1}{2(1-a)}\left(1 - \left(1-x^2\right)^{1-a}\right)$. As $x \to 1$ the limit is $\frac{1}{2(1-a)}$ if $a < 1$, and the limit diverges to $+\infty$ if $a \geq 1$.

317. $\int_{t=\pi}^{0} b\sqrt{1-\cos^2 t} \times (-a\sin t)dt = \int_{t=0}^{\pi} ab\sin^2 t dt$

319. $f(t) = 2\cos(3t) - \cos(2t)$; $\int_0^\pi (2\cos(3t) - \cos(2t))\,dt = 0$

Section 5.6 Exercises

321. $\frac{-1}{3}e^{-3x} + C$

323. $-\frac{3^{-x}}{\ln 3} + C$

325. $2\ln|x| + C$ or $\ln\left(x^2\right) + C$

327. $2\sqrt{x} + C$

329. $-\frac{1}{\ln x}+C$

331. $\ln(\ln(\ln x))+C$

333. $\ln(x\cos x)+C$

335. $-\frac{1}{2}(\ln(\cos(x)))^2+C$

337. $\frac{-e^{-x^3}}{3}+C$

339. $e^{\tan x}+C$

341. $t+C$

343. $\frac{2}{9}x^3\left(\ln\left(x^3\right)-1\right)+C$

345. $2\sqrt{x}(\ln x-2)+C$

347. $\int_0^{\ln x} e^t\,dt = e^t\Big|_0^{\ln x} = e^{\ln x}-e^0 = x-1$

349. $-\frac{1}{3}\ln|\sin(3x)+\cos(3x)|+\mathrm{C}$

351. $-\frac{1}{2}\ln\left|\csc\left(x^2\right)+\cot\left(x^2\right)\right|+C$

353. $-\frac{1}{2}(\ln(\csc x))^2+C$

355. $\frac{1}{3}\ln\left(\frac{26}{7}\right)$

357. $\ln\left(\sqrt{3}-1\right)$

359. $\frac{1}{2}\ln\frac{3}{2}$

361. $y-2\ln|y+1|+C$

363. $\ln|\sin x-\cos x|+C$

365. $-\frac{1}{3}\left(1-\left(\ln x\right)^2\right)^{3/2}+C$

367. Exact solution: $\frac{e-1}{e}$, $R_{50}=0.6258$. Since f is decreasing, the right endpoint estimate underestimates the area.

369. Exact solution: $\frac{2\ln(3)-\ln(6)}{2}$, $R_{50}=0.2033$. Since f is increasing, the right endpoint estimate overestimates the area.

371. Exact solution: $-\frac{1}{\ln(4)}$, $R_{50}=-0.7164$. Since f is negative and increasing on the interval, the right-endpoint estimate overestimates the value of the integral, but it underestimates the area of the region (absolute value of the integral).

373. $\frac{11}{2}\ln 2$

375. $\frac{1}{\ln(65{,}536)}$

377. $\int_N^{N+1} xe^{-x^2}\,dx = \frac{1}{2}\left(e^{-N^2}-e^{-(N+1)^2}\right)$. The quantity is less than 0.01 when $N=2$.

379. $\int_a^b \frac{dx}{x} = \ln(b)-\ln(a) = \ln\left(\frac{1}{a}\right)-\ln\left(\frac{1}{b}\right) = \int_{1/b}^{1/a}\frac{dx}{x}$

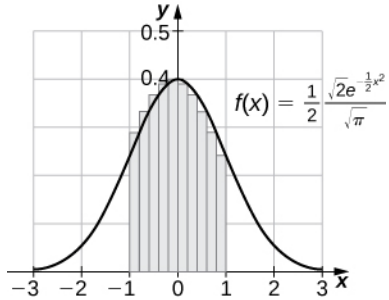
381. 23

383. We may assume that $x>1$, so $\frac{1}{x}<1$. Then, $\int_1^{1/x}\frac{dt}{t}$. Now make the substitution $u=\frac{1}{t}$, so $du=-\frac{dt}{t^2}$ and $\frac{du}{u}=-\frac{dt}{t}$, and change endpoints: $\int_1^{1/x}\frac{dt}{t}=-\int_1^x\frac{du}{u}=-\ln x$.

385. Answers will vary.

387. $x=E(\ln(x))$. Then, $1=\frac{E'(\ln x)}{x}$ or $x=E'(\ln x)$. Since any number t can be written $t=\ln x$ for some x, and for such t we have $x=E(t)$, it follows that for any t, $E'(t)=E(t)$.

389. $R_{10}=0.6811$, $R_{100}=0.6827$



Section 5.7 Exercises

391. $\sin^{-1} x\Big|_0^{\sqrt{3}/2} = \frac{\pi}{3}$

393. $\tan^{-1} x\Big|_{\sqrt{3}}^{1} = -\frac{\pi}{12}$

395. $\sec^{-1} |_{\frac{\sqrt{2}}{\sqrt{3}}}^{\frac{2}{}} = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$

397. $\sin^{-1}\left(\frac{x}{3}\right) + C$

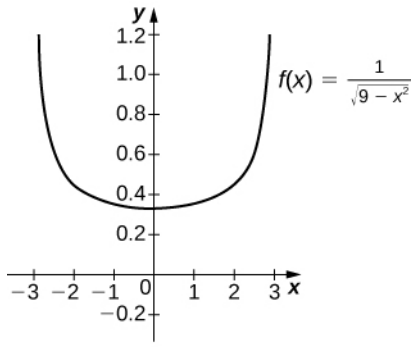
399. $\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$

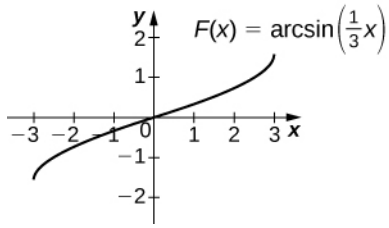
401. $\frac{1}{3}\sec^{-1}\left(\frac{x}{3}\right) + C$

403. $\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$. So, $\sin^{-1} t = \frac{\pi}{2} - \cos^{-1} t$. They differ by a constant.

405. $\sqrt{1 - t^2}$ is not defined as a real number when $t > 1$.

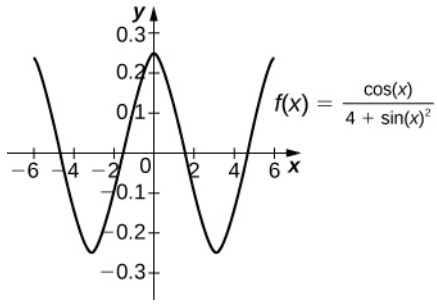
407.

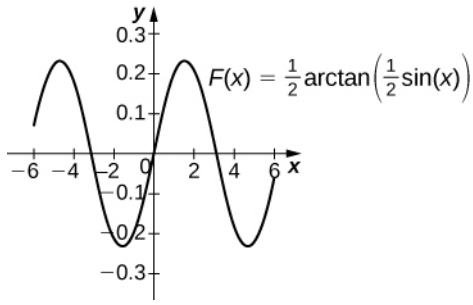




The antiderivative is $\sin^{-1}\left(\frac{x}{3}\right) + C$. Taking $C = \frac{\pi}{2}$ recovers the definite integral.

409.





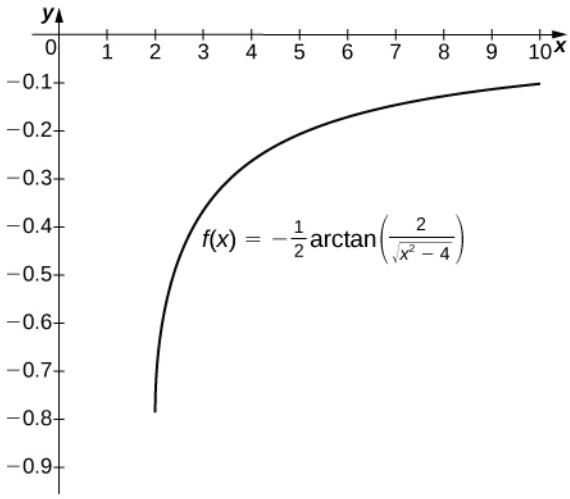
The antiderivative is $\frac{1}{2}\tan^{-1}\left(\frac{\sin x}{2}\right) + C$. Taking $C = \frac{1}{2}\tan^{-1}\left(\frac{\sin(6)}{2}\right)$ recovers the definite integral.

411. $\frac{1}{2}\left(\sin^{-1} t\right)^2 + C$

413. $\frac{1}{4}\left(\tan^{-1}(2t)\right)^2 + C$

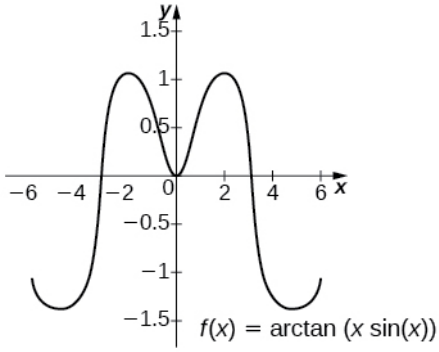
415. $\frac{1}{4}\left(\sec^{-1}\left(\frac{t}{2}\right)^2\right) + C$

417.



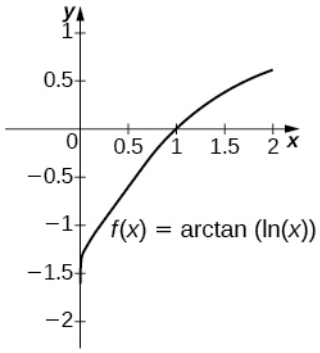
The antiderivative is $\frac{1}{2}\sec^{-1}\left(\frac{x}{2}\right) + C$. Taking $C = 0$ recovers the definite integral over $[2, 6]$.

419.



The general antiderivative is $\tan^{-1}(x\sin x)+C$. Taking $C=-\tan^{-1}(6\sin(6))$ recovers the definite integral.

421.



The general antiderivative is $\tan^{-1}(\ln x)+C$. Taking $C=\frac{\pi}{2}=\tan^{-1}\infty$ recovers the definite integral.

423. $\sin^{-1}\left(e^t\right)+C$

425. $\sin^{-1}(\ln t)+C$

427. $-\frac{1}{4}\left(\cos^{-1}(2t)\right)^2+C$

429. $\frac{1}{2}\ln\left(\frac{4}{3}\right)$

431. $1-\frac{2}{\sqrt{5}}$

433. $2\tan^{-1}(A)\to\pi$ as $A\to\infty$

435. Using the hint, one has $\int\frac{\csc^2 x}{\csc^2 x+\cot^2 x}dx=\int\frac{\csc^2 x}{1+2\cot^2 x}dx$. Set $u=\sqrt{2}\cot x$. Then, $du=-\sqrt{2}\csc^2 x$ and the integral is $-\frac{1}{\sqrt{2}}\int\frac{du}{1+u^2}=-\frac{1}{\sqrt{2}}\tan^{-1}u+C=\frac{1}{\sqrt{2}}\tan^{-1}\left(\sqrt{2}\cot x\right)+C$. If one uses the identity $\tan^{-1}s+\tan^{-1}\left(\frac{1}{s}\right)=\frac{\pi}{2}$, then this can also be written $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan x}{\sqrt{2}}\right)+C$.

437. $x\approx\pm1.7321$. The left endpoint estimate with $N=100$ is 4.781 and these decimals persist for $N=500$.

Review Exercises

439. False

441. True

443. $L_4=5.25$, $R_4=3.25$, exact answer: 4

445. $L_4=5.364$, $R_4=5.364$, exact answer: 5.870

447. $-\frac{4}{3}$

449. 1

451. $-\frac{1}{2(x+4)^2}+C$

453. $\frac{4}{3}\sin^{-1}\left(x^3\right)+C$

455. $\frac{\sin t}{\sqrt{1+t^2}}$
457. $4\frac{\ln x}{x}+1$
459. $6,328,113
461. $73.36
463. $\frac{19117}{12}$ ft/sec, or 1593 ft/sec

Chapter 6

Checkpoint

6.1 12 units^2
6.2 $\frac{3}{10}$ unit^2
6.3 $2+2\sqrt{2}$ units^2
6.4 $\frac{5}{3}$ units^2
6.5 $\frac{5}{3}$ units^2
6.7 $\frac{\pi}{2}$
6.8 8π units^3
6.9 21π units^3
6.10 $\frac{10\pi}{3}$ units^3
6.11 60π units^3
6.12 $\frac{15\pi}{2}$ units^3
6.13 8π units^3
6.14 12π units^3
6.15 $\frac{11\pi}{6}$ units^3
6.16 $\frac{\pi}{6}$ units^3
6.17 Use the method of washers; $V=\int_{-1}^{1}\pi\left[\left(2-x^2\right)^2-\left(x^2\right)^2\right]dx$
6.18 $\frac{1}{6}\left(5\sqrt{5}-1\right)\approx 1.697$
6.19 Arc Length ≈ 3.8202
6.20 Arc Length $=3.15018$
6.21 $\frac{\pi}{6}\left(5\sqrt{5}-3\sqrt{3}\right)\approx 3.133$
6.22 12π
6.23 $70/3$
6.24 24π
6.25 8 ft-lb
6.26 Approximately 43,255.2 ft-lb
6.27 156,800 N
6.28 Approximately 7,164,520,000 lb or 3,582,260 t
6.29 $M=24, \overline{x}=\frac{2}{5}$ m
6.30 $(-1,-1)$ m
6.31 The centroid of the region is $(3/2, 6/5)$.
6.32 The centroid of the region is $(1, 13/5)$.
6.33 The centroid of the region is $(0, 2/5)$.
6.34 $6\pi^2$ units^3
6.35 a. $\frac{d}{dx}\ln\left(2x^2+x\right)=\frac{4x+1}{2x^2+x}$
b. $\frac{d}{dx}\left(\ln\left(x^3\right)\right)^2=\frac{6\ln\left(x^3\right)}{x}$
6.36 $\int\frac{x^2}{x^3+6}dx=\frac{1}{3}\ln\left|x^3+6\right|+C$
6.37 $4\ln 2$
6.38 a. $\frac{d}{dx}\left(\frac{e^{x^2}}{e^{5x}}\right)=e^{x^2-5x}(2x-5)$

b. $\frac{d}{dt}\left(e^{2t}\right)^3 = 6e^{6t}$

6.39 $\int \frac{4}{e^{3x}}dx = -\frac{4}{3}e^{-3x} + C$

6.40 a. $\frac{d}{dt}4^{t^4} = 4^{t^4}\,(\ln 4)\left(4t^3\right)$

b. $\frac{d}{dx}\log_3\left(\sqrt{x^2+1}\right) = \frac{x}{(\ln 3)\left(x^2+1\right)}$

6.41 $\int x^2 2^{x^3}\,dx = \frac{1}{3\ln 2}2^{x^3} + C$

6.42 There are 81,377,396 bacteria in the population after 4 hours. The population reaches 100 million bacteria after 244.12 minutes.

6.43 At 5% interest, she must invest \$223,130.16. At 6% interest, she must invest \$165,298.89.

6.44 38.90 months

6.45 The coffee is first cool enough to serve about 3.5 minutes after it is poured. The coffee is too cold to serve about 7 minutes after it is poured.

6.46 A total of 94.13 g of carbon remains. The artifact is approximately 13,300 years old.

6.47 a. $\frac{d}{dx}\left(\tanh\left(x^2+3x\right)\right) = \left(\operatorname{sech}^2\left(x^2+3x\right)\right)(2x+3)$

b. $\frac{d}{dx}\left(\frac{1}{(\sinh x)^2}\right) = \frac{d}{dx}(\sinh x)^{-2} = -2(\sinh x)^{-3}\cosh x$

6.48 a. $\int \sinh^3 x\cosh x\,dx = \frac{\sinh^4 x}{4} + C$

b. $\int \operatorname{sech}^2(3x)\,dx = \frac{\tanh(3x)}{3} + C$

6.49 a. $\frac{d}{dx}\left(\cosh^{-1}(3x)\right) = \frac{3}{\sqrt{9x^2-1}}$

b. $\frac{d}{dx}\left(\coth^{-1}x\right)^3 = \frac{3\left(\coth^{-1}x\right)^2}{1-x^2}$

6.50 a. $\int \frac{1}{\sqrt{x^2-4}}dx = \cosh^{-1}\left(\frac{x}{2}\right) + C$

b. $\int \frac{1}{\sqrt{1-e^{2x}}}dx = -\operatorname{sech}^{-1}(e^x) + C$

6.51 52.95 ft

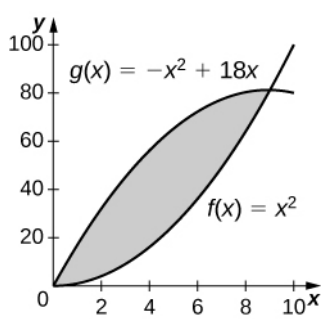
Section 6.1 Exercises

1. $\frac{32}{3}$

3. $\frac{13}{12}$

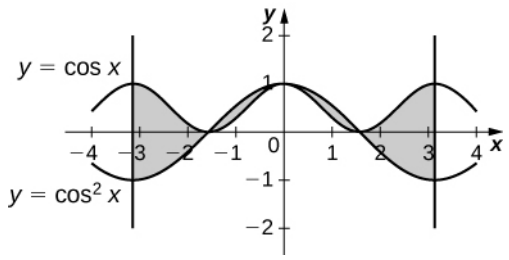
5. 36

7.



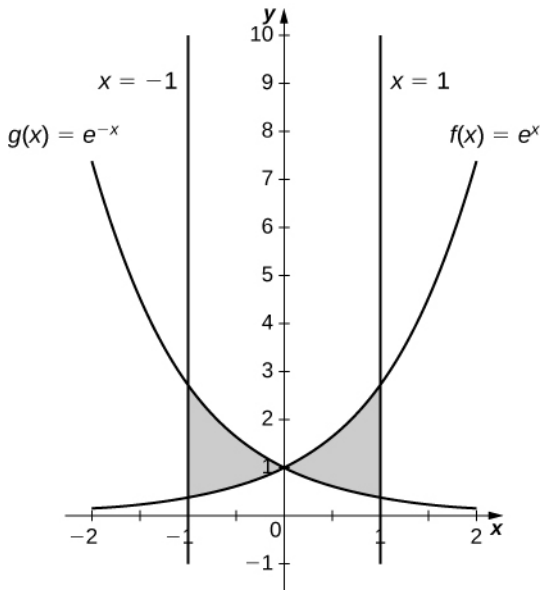
243 square units

9.



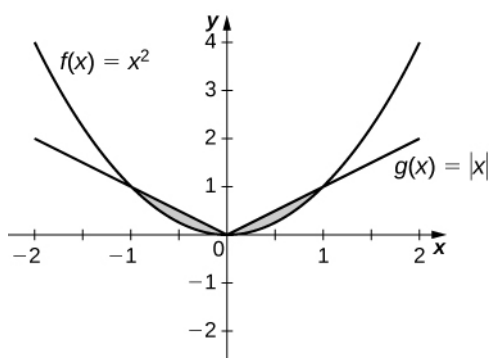
4

11.



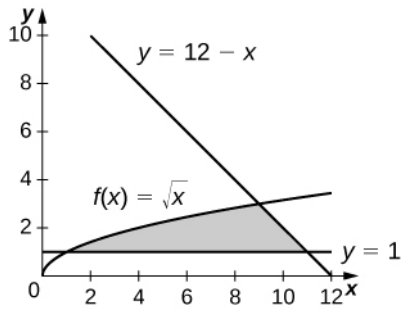
$\frac{2(e-1)^2}{e}$

13.



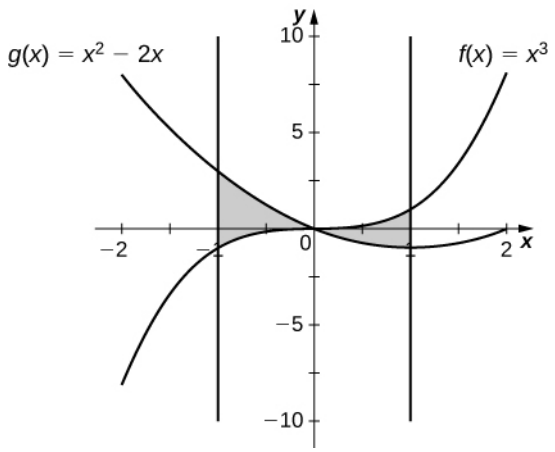
$\frac{1}{3}$

15.



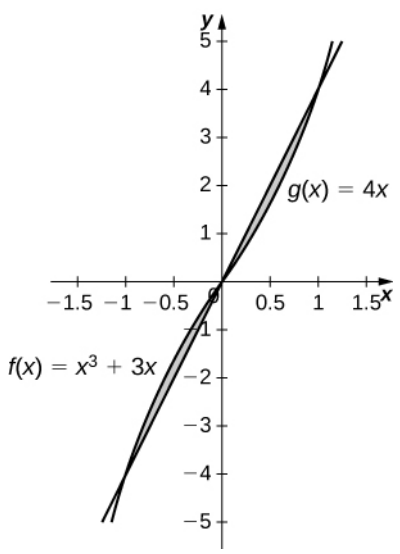
$\frac{34}{3}$

17.



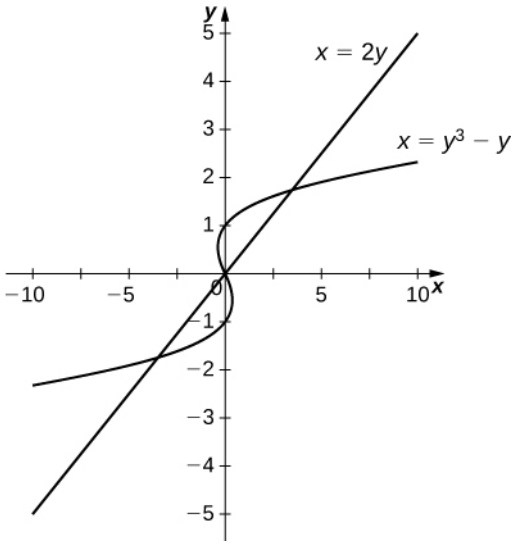
$\frac{5}{2}$

19.



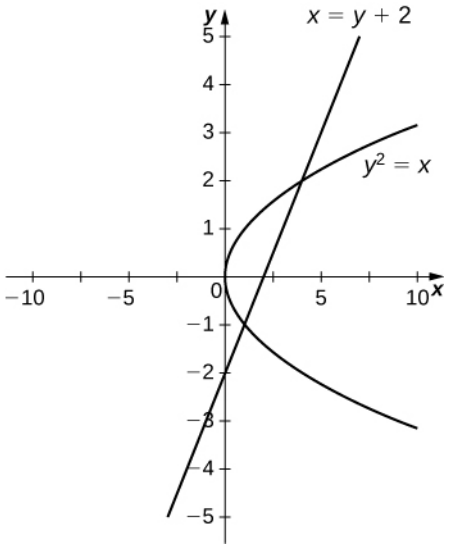
$\frac{1}{2}$

21.



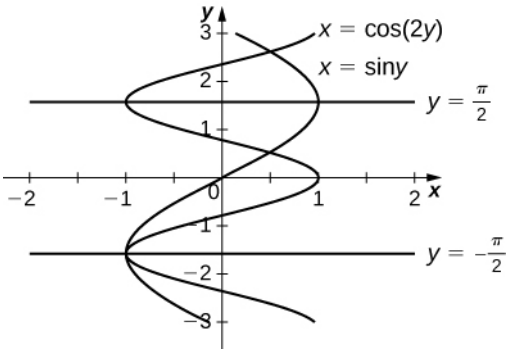
$\frac{9}{2}$

23.



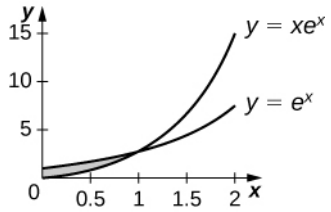
$\frac{9}{2}$

25.



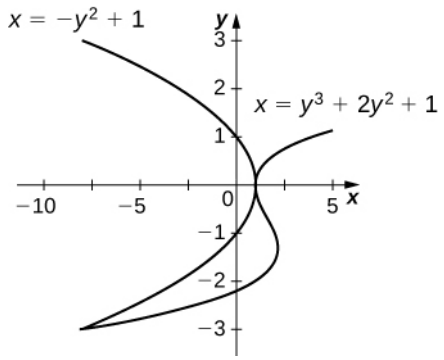
$\frac{3\sqrt{3}}{2}$

27.



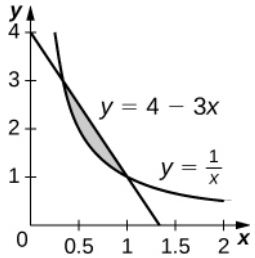
$e-2$

29.



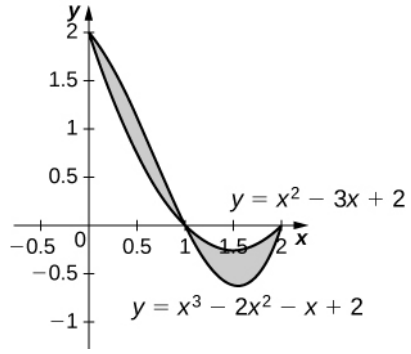
$\frac{27}{4}$

31.



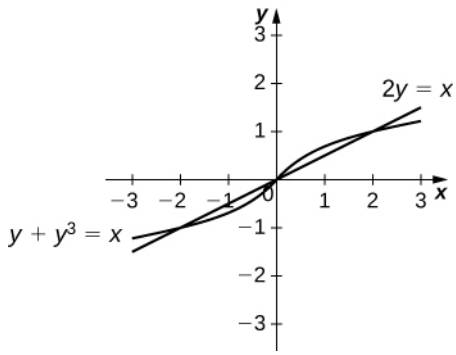
$\frac{4}{3} - \ln(3)$

33.



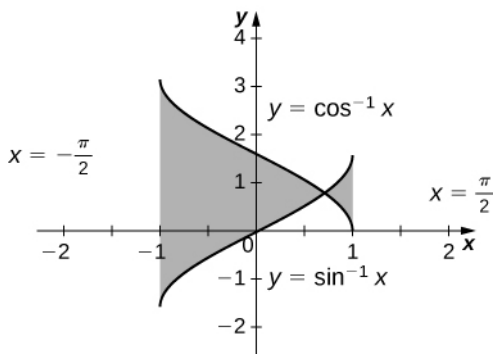
$\frac{1}{2}$

35.



$\frac{1}{2}$

37.



$-2\left(\sqrt{2}-\pi\right)$

39. 1.067

41. 1.605

43. 7.523

45. $\frac{3\pi-4}{12}$

47. 1.429

49. $\$33{,}333.33$ total profit for 200 cell phones sold

51. 3.32 mi represents how far ahead the hare is from the tortoise

53. $\frac{343}{24}$

55. $4\sqrt{3}$

57. $\pi-\frac{32}{25}$

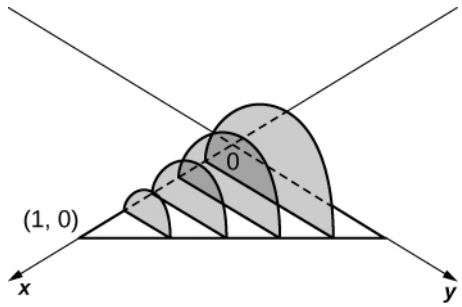
Section 6.2 Exercises

63. 8 units^3

65. $\frac{32}{3\sqrt{2}}$ units^3

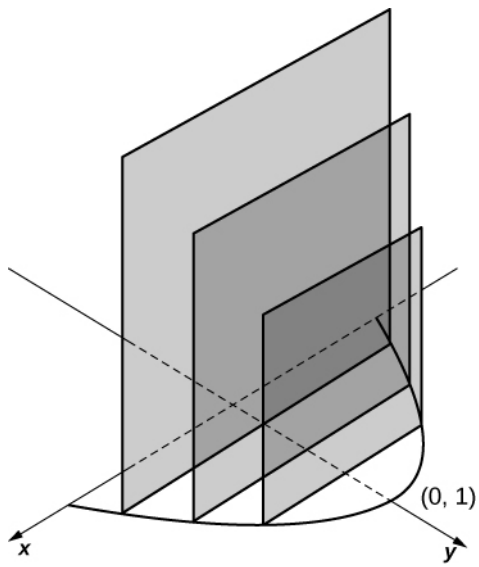
67. $\frac{7}{24}\pi r^2 h$ units^3

69.



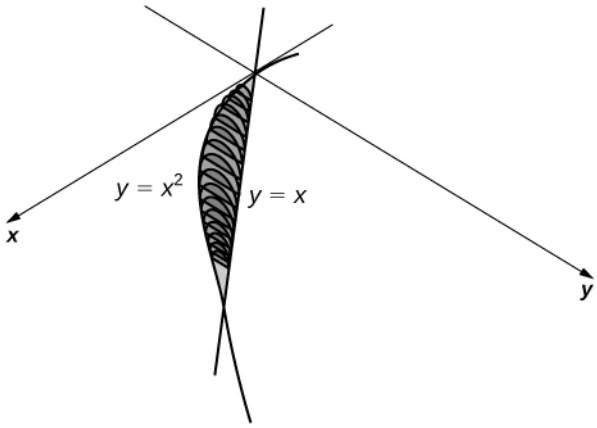
$\frac{\pi}{24}$ units^3

71.



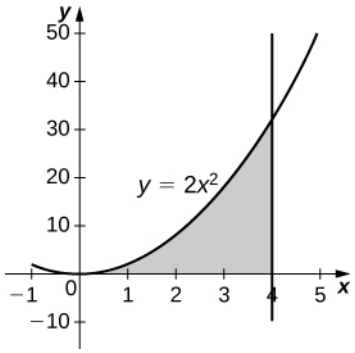
2 units^3

73.



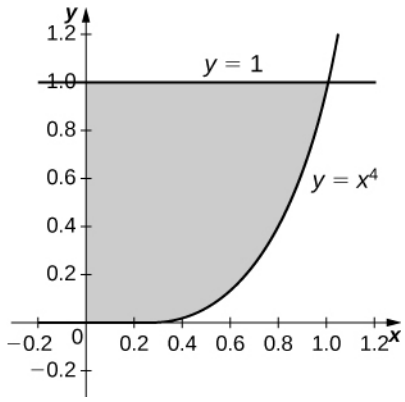
$\frac{\pi}{240}$ units^3

75.



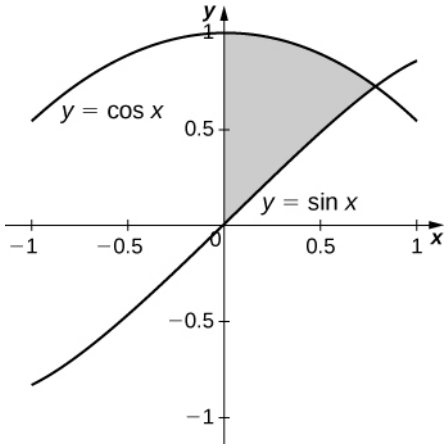
$\frac{4096\pi}{5}$ units^3

77.



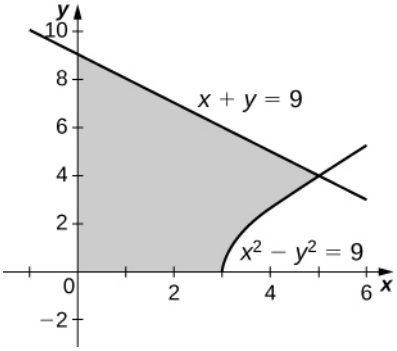
$\frac{8\pi}{9}$ units^3

79.



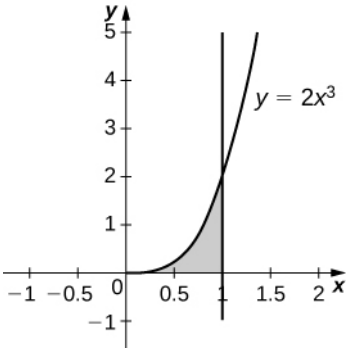
$\frac{\pi}{2}$ units^3

81.



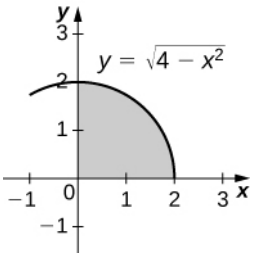
207π units^3

83.



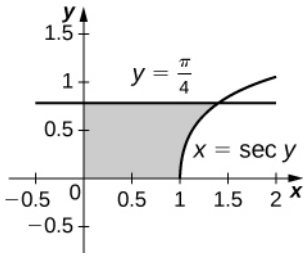
$\frac{4\pi}{5}$ units^3

85.



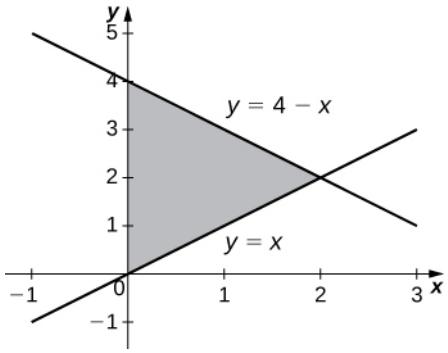
$\frac{16\pi}{3}$ units^3

87.



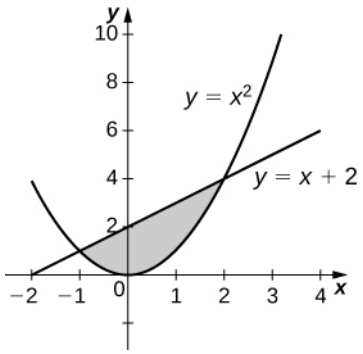
π units^3

89.



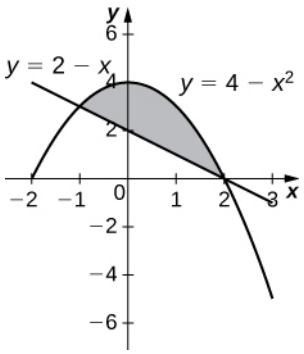
$\frac{16\pi}{3}$ units^3

91.



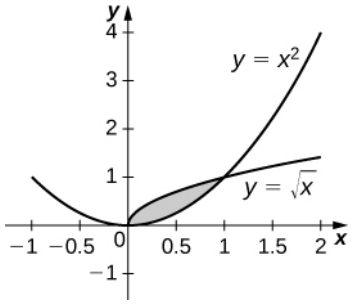
$\frac{72\pi}{5}$ units^3

93.



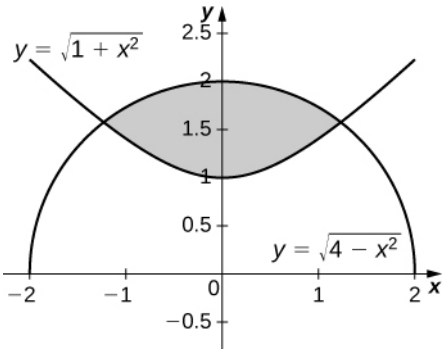
$\frac{108\pi}{5}$ units^3

95.



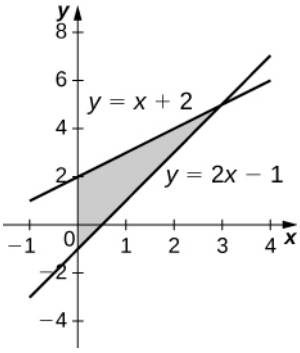
$\frac{3\pi}{10}$ units^3

97.



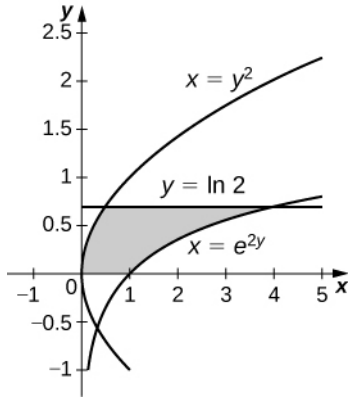
$2\sqrt{6}\pi$ units^3

99.



9π units^3

101.



$\frac{\pi}{20}\left(75 - 4\ln^5(2)\right)$ units^3

103. $\frac{m^2\pi}{3}\left(b^3 - a^3\right)$ units^3

105. $\frac{4a^2b\pi}{3}$ units^3

107. $2\pi^2$ units^3

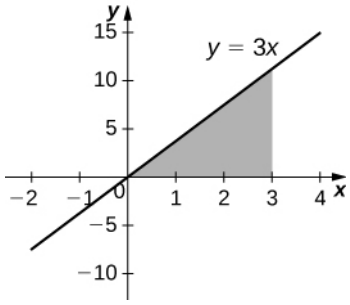
109. $\frac{2ab^2\pi}{3}$ units^3

111. $V = \frac{2\pi}{3}(2r + h)(r - h)^2$ units^3

113. $\frac{\pi}{3}(h + R)(h - 2R)^2$ units^3

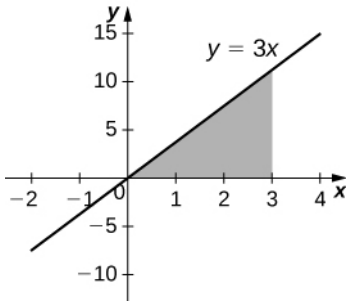
Section 6.3 Exercises

115.



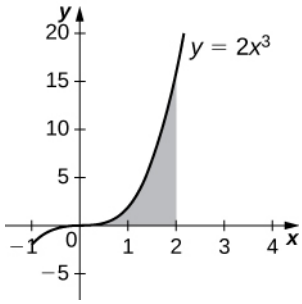
54π units^3

117.



81π units^3

119.



$\frac{512\pi}{7}$ units^3

121. 2π units^3

123. $\frac{2\pi}{3}$ units^3

125. 2π units^3

127. $\frac{4\pi}{5}$ units^3

129. $\frac{64\pi}{3}$ units^3

131. $\frac{32\pi}{5}$ units^3

133. $\frac{7\pi}{6}$

135. 48π

137. $\frac{308\pi}{5}$

139. $\frac{512\pi}{7}$

141. $\frac{8\pi}{5}$

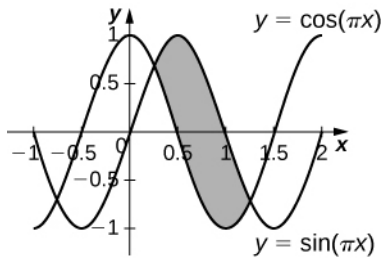
143. $\frac{28\pi}{15}$ units^3

145. $\frac{3\pi}{10}$ units^3

147. $\pi\left(\frac{6\cdot 2^{2/3}}{5}-\frac{11}{10}\right)=\frac{\pi}{10}\left(12\cdot 2^{2/3}-11\right)\approx 2.5286$ units^3

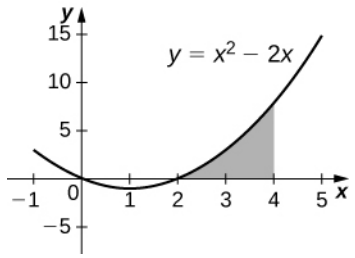
149. 0.9876 units^3

151.



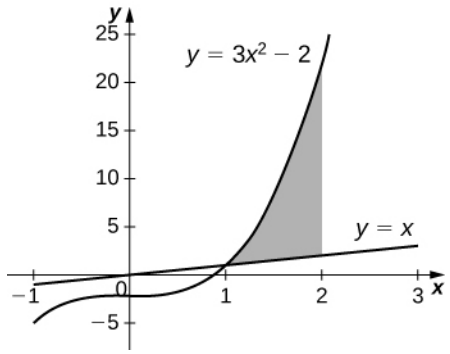
$3\sqrt{2}$ units^3

153.



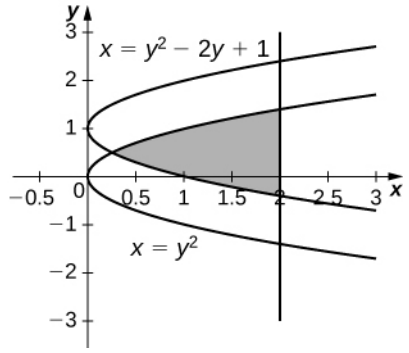
$\frac{496\pi}{15}$ units^3

155.



$\frac{398\pi}{15}$ units^3

157.



15.9074 units^3

159. $\frac{1}{3}\pi r^2 h$ units^3
161. $\pi r^2 h$ units^3
163. πa^2 units^3

Section 6.4 Exercises

165. $2\sqrt{26}$
167. $2\sqrt{17}$
169. $\frac{\pi}{6}\left(17\sqrt{17}-5\sqrt{5}\right)$
171. $\frac{13\sqrt{13}-8}{27}$
173. $\frac{4}{3}$
175. 2.0035
177. $\frac{123}{32}$
179. 10
181. $\frac{20}{3}$
183. $\frac{1}{675}\left(229\sqrt{229}-8\right)$
185. $\frac{1}{8}\left(4\sqrt{5}+\ln\left(9+4\sqrt{5}\right)\right)$
187. 1.201
189. 15.2341
191. $\frac{49\pi}{3}$
193. $70\pi\sqrt{2}$
195. 8π
197. $120\pi\sqrt{26}$
199. $\frac{\pi}{6}\left(17\sqrt{17}-1\right)$
201. $9\sqrt{2}\pi$
203. $\frac{10\sqrt{10}\pi}{27}\left(73\sqrt{73}-1\right)$
205. 25.645
207. $\pi(2+\pi)$
209. 10.5017
211. 24 ft
213. 2
215. Answers may vary
217. For more information, look up Gabriel's Horn.

Section 6.5 Exercises

219. 150 ft-lb
221. 200 J
223. 1 J
225. $\frac{39}{2}$
227. $\ln(243)$
229. $\frac{332\pi}{15}$
231. 100π
233. $20\pi\sqrt{15}$
235. 6 J
237. 5 cm
239. 36 J
241. $18{,}750$ ft-lb
243. Weight $= \frac{32}{3} \times 10^9$ lb
Work $= \frac{4}{3} \times 10^{12}$ ft-lb
245. 9.71×10^8 N m
247. a. $3{,}000{,}000$ lb, b. $750{,}000$ lbs

249. 23.25π million ft-lb

251. $\frac{A\rho H^2}{2}$

253. Answers may vary

Section 6.6 Exercises

255. $\frac{5}{4}$

257. $\left(\frac{2}{3}, \frac{2}{3}\right)$

259. $\left(\frac{7}{4}, \frac{3}{2}\right)$

261. $\frac{3L}{4}$

263. $\frac{\pi}{2}$

265. $\frac{e^2+1}{e^2-1}$

267. $\frac{\pi^2-4}{\pi}$

269. $\frac{1}{4}\left(1+e^2\right)$

271. $\left(\frac{a}{3}, \frac{b}{3}\right)$

273. $\left(0, \frac{\pi}{8}\right)$

275. $(0, 3)$

277. $\left(0, \frac{4}{\pi}\right)$

279. $\left(\frac{5}{8}, \frac{1}{3}\right)$

281. $\frac{2m\pi}{3}$

283. $\pi a^2 b$

285. $\left(\frac{4}{3\pi}, \frac{4}{3\pi}\right)$

287. $\left(\frac{1}{2}, \frac{2}{5}\right)$

289. $\left(0, \frac{28}{9\pi}\right)$

291. Center of mass: $\left(\frac{3a}{8}, \frac{2a^2}{5}\right)$, volume: $\frac{\pi a^4}{2}$

293. Volume: $2\pi^2 a^2 (b+a)$

Section 6.7 Exercises

295. $\frac{1}{x}$

297. $-\frac{1}{x(\ln x)^2}$

299. $\ln(x+1)+C$

301. $\ln(x)+1$

303. $\cot(x)$

305. $\frac{7}{x}$

307. $\csc(x)\sec x$

309. $-2\tan x$

311. $\frac{1}{2}\ln\left(\frac{5}{3}\right)$

313. $2-\frac{1}{2}\ln(5)$

315. $\frac{1}{\ln(2)}-1$

317. $\frac{1}{2}\ln(2)$

319. $\frac{1}{3}(\ln x)^3$

321. $\frac{2x^3}{\sqrt{x^2+1}\sqrt{x^2-1}}$

323. $x^{-2-(1/x)}(\ln x-1)$

325. ex^{e-1}

327. 1

329. $-\frac{1}{x^2}$

331. $\pi-\ln(2)$

333. $\frac{1}{x}$

335. e^5-6 units^2

337. $\ln(4) - 1$ units^2
339. 2.8656
341. 3.1502

Section 6.8 Exercises

349. True
351. True; $k = \frac{\ln(2)}{t}$
353. 20 hours
355. No. The relic is approximately 871 years old.
357. 73 years
359. 5 days 6 hours 27 minutes
361. 12
363. 8.618%
365. \$6766.76
367. 9 hours 13 minutes
369. 239,179 years
371. $P'(t) = 43e^{0.01604t}$. The population is always increasing.
373. The population reaches 10 billion people in 2032.
375. $P'(t) = 2.259e^{0.06407t}$. The population is always increasing.

Section 6.9 Exercises

377. e^x and e^{-x}
379. Answers may vary
381. Answers may vary
383. Answers may vary
385. $3\sinh(3x+1)$
387. $-\tanh(x)\operatorname{sech}(x)$
389. $4\cosh(x)\sinh(x)$
391. $\frac{x\operatorname{sech}^2\left(\sqrt{x^2+1}\right)}{\sqrt{x^2+1}}$
393. $6\sinh^5(x)\cosh(x)$
395. $\frac{1}{2}\sinh(2x+1) + C$
397. $\frac{1}{2}\sinh\left(x^2\right) + C$
399. $\frac{1}{3}\cosh^3(x) + C$
401. $\ln(1+\cosh(x)) + C$
403. $\cosh(x) + \sinh(x) + C$
405. $\frac{4}{1-16x^2}$
407. $\frac{\sinh(x)}{\sqrt{\cosh^2(x)+1}}$
409. $-\csc(x)$
411. $-\frac{1}{\left(x^2-1\right)\tanh^{-1}(x)}$
413. $\frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C$
415. $\sqrt{x^2+1} + C$
417. $\cosh^{-1}(e^x) + C$
419. Answers may vary
421. 37.30
423. $y = \frac{1}{c}\cosh(cx)$
425. -0.521095
427. 10

Review Exercises

435. False
437. False

439. $16\sqrt{3}$ units^3

441. $\frac{162\pi}{5}$

443. a. 4, b. $\frac{128\pi}{7}$, c. $\frac{64\pi}{5}$

445. a. 1.949, b. 21.952, c. 17.099

447. a. $\frac{31}{6}$, b. $\frac{452\pi}{15}$, c. $\frac{31\pi}{6}$

449. $\pi \ln 17$

451. Mass: $\frac{1}{2}$, center of mass: $\left(\frac{18}{35}, \frac{9}{22}\right)$

453. $\sqrt{17} + \frac{1}{8}\ln(33 + 8\sqrt{17})$

455. Volume: $\frac{3\pi}{4}$, surface area: $\pi\left(\sqrt{2} - \sinh^{-1}(1) + \sinh^{-1}(16) - \frac{\sqrt{257}}{16}\right)$

457. 11:02 a.m.

459. $\pi(1 + \sinh(1)\cosh(1))$

Index

A
absolute extremum 318
absolute maximum 318
absolute minimum 318
absolute value function 23
acceleration 197, 230
algebraic function 40
amount of change 231
antiderivative 420
aphelion 487
arc length 582
Archimedes 152, 438
area density 596
Area Problem 110
area under the curve 448
average rate of change 231
average value of the function 467
average velocity 109, 195

B
bald eagle 502
base 80

C
carbon dating 642
catenary 652
center of mass 611
centroid 614
chain rule 250
change of variables 507
chaos 415
common logarithm 87
composite function 19
compound interest 638
compounding interest 83
concave down 344
concave up 344
concavity 344
concavity test 344
conditional statement 169
constant function 34
Constant multiple law for limits 141
Constant Multiple Rule 219
constant rule 216
continuity over an interval 162
continuous at a point 157
continuous from the left 162
continuous from the right 162
critical number 321
cross-section 551
cubic function 34

D
deceleration 501
decreasing on the interval I 17
definite integral 456
degree 34
density function 595
dependent variable 8
derivative 193
derivative function 203
Difference law for limits 141
difference quotient 189
Difference Rule 219
differentiable at a 203
differentiable function 203
differentiable on S 203
Differential calculus 108
differential form 312
differentials 312
differentiation 193
discontinuous at a point 157
disk method 557
displacement 461, 489
domain 8
doubling time 640
dummy variable 438, 456

E
earthquake 88
end behavior 35, 361
endpoints 10
epsilon-delta definition of the limit 169
evaluation theorem 478
even function 22, 493
existential quantifier 169
exponent 80
exponential decay 640
exponential growth 637
Extreme Value Theorem 318

F
f_{ave} 467
federal income tax 502
Fermat's theorem 321
first derivative test 341
folium of Descartes 273
fruit flies 520
frustum 586
function 8
Fundamental Theorem of Calculus 473
Fundamental Theorem of Calculus, Part 1 475
Fundamental Theorem of Calculus, Part 2 478

G
graph of a function 9
growth of bacteria 519

H
half-life 642
hanging cables 652
higher-order derivatives 210
Holling type I equation 240
Hooke's law 599
Hoover Dam 606
horizontal asymptote 355
horizontal line test 66
hydrostatic pressure 603
hyperbolic functions 89

I
iceboat 492
implicit differentiation 269
increasing on the interval I 17
indefinite integral 421
independent variable 8
indeterminate forms 395
index 438
infinite discontinuity 160
infinite limit at infinity 360
infinite limits 126
inflection point 345
initial-value problem 427
input 8
instantaneous rate of change 196
instantaneous velocity 109, 195
integrable function 456
Integral calculus 111
integrand 456
integration by substitution 506
interior points 215
Intermediate Value Theorem 164
interval notation 10
intuitive definition of the limit 117
inverse function 66
inverse hyperbolic functions 93
inverse trigonometric functions 72
iterative process 412

J
joule 597
jump discontinuity 160

L
L'Hôpital's rule 394
lamina 614
leading coefficient 34
left-endpoint approximation 442
Leibniz 188, 456
limit 109
limit at infinity 355, 359
limit laws 141
limits of integration 456
linear approximation 309

linear function 30
linearization 309
local extremum 320
local maximum 320
local minimum 320
logarithmic differentiation 282
logarithmic function 42
lower sum 449

M

Mandelbrot set 415
marginal cost 234
marginal profit 234
mathematical models 38
maximizing revenue 386
Mean Value Theorem 329
Mean Value Theorem for Integrals 473
method of cylindrical shells. 572
method of exhaustion 438
moment 611
multivariable calculus 112

N

natural exponential function 84, 276
natural logarithm 85
natural logarithmic function 276
net change theorem 489
net signed area 460
Newton 188, 473
Newton's law of cooling 641
Newton's method 409
number e 84

O

oblique asymptote 366
odd function 23, 493
one-sided limit 123
one-to-one function 66
optimization problems 381
output 8

P

partition 442
Pascal's principle 604
pascals 603
percentage error 315
perihelion 487
periodic functions. 60
piecewise-defined function 43
piecewise-defined functions 10
point-slope equation 32
polynomial function 34
population growth 80, 637
population growth rates 230
power function 34
Power law for limits 141
power rule 218
price–demand function 517
Product law for limits 141
product rule 221
propagated error 314
Pythagorean theorem 298

Q

quadratic function 34
Quotient law for limits 141
quotient rule 223

R

radial density 596
radians 53
range 8
rate of change 106, 489
rational function 40
Regiomontanus' problem 435
regular partition 442
related rates 296
relative error 315
removable discontinuity 160
restricted domain 71
Richter scale 88
Riemann sum 448
right-endpoint approximation 443
Rolle's theorem 329
root function 40
Root law for limits 141

S

secant 107
secant method 419
second derivative test 347
sigma notation 438
simple interest 638
skydiver 482
slicing method 553
slope 31
slope-intercept form 32
smooth 582
solid of revolution 554
speed 232
spring constant 599
squeeze theorem 148
standard form of a line 32
Sum law for limits 141
Sum Rule 219
summation notation 438
sums and powers of integers 440
Surface area 585
symmetry about the origin 22
symmetry about the y-axis 22
symmetry principle 614

T

table of values 12
tangent 108
tangent line approximation 309
Tangent Problem 108
theorem of Pappus for volume 622
total area 462
Tour de France 499
transcendental functions 42
transformation of a function 45
triangle inequality 175
trigonometric functions 55
trigonometric identity 58

U

universal quantifier 169
upper sum 449

V

variable of integration 456
velocity 489
vertical asymptote 128
vertical line test 14

W

washer method 562
wingsuits 483
work 598

Z

zeroes of functions 409
zeros of a function 14